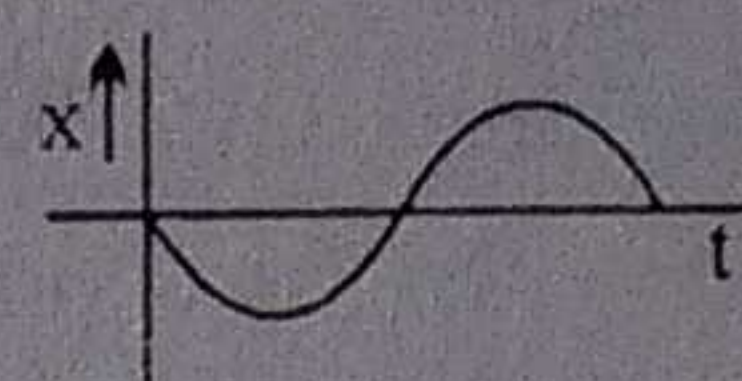


ONLY ONE OPTION IS CORRECT.

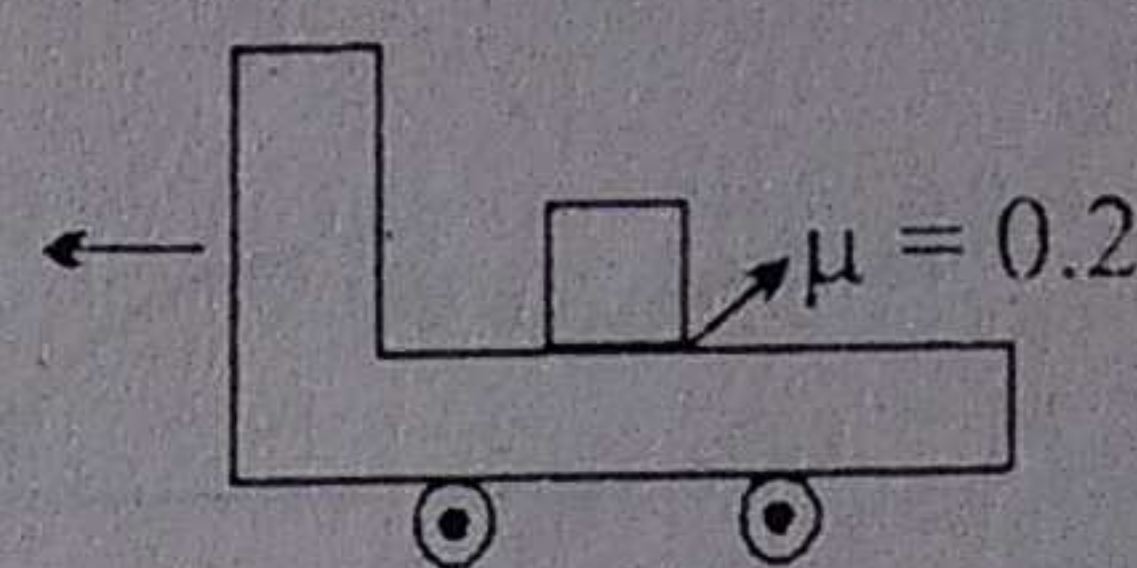
Take approx. 2 minutes for answering each question.

Q.1 If position time graph of a particle is sine curve as shown, what will be its velocity-time graph.



- (A) (B) (C) (D)

Q.2 A truck starting from rest moves with an acceleration of 5 m/s^2 for 1 sec and then moves with constant velocity. The velocity w.r.t ground v/s time graph for block in truck is (Assume that block does not fall off the truck)



- (A) (B) (C) (D) None of these

Q.3 If angular velocity of a disc depends on an angle rotated θ as $\omega = \theta^2 + 2\theta$, then its angular acceleration α at $\theta = 1 \text{ rad}$ is :

- (A) 8 rad/sec^2 (B) 10 rad/sec^2 (C) 12 rad/sec^2 (D) None

Q.4 If a particle takes t second less and acquires a velocity of $v \text{ ms}^{-1}$ more in falling through the same distance (starting from rest) on two planets where the accelerations due to gravity are $2g$ and $8g$ respectively then:

- (A) $v = 2gt$ (B) $v = 4gt$ (C) $v = 5gt$ (D) $v = 16gt$

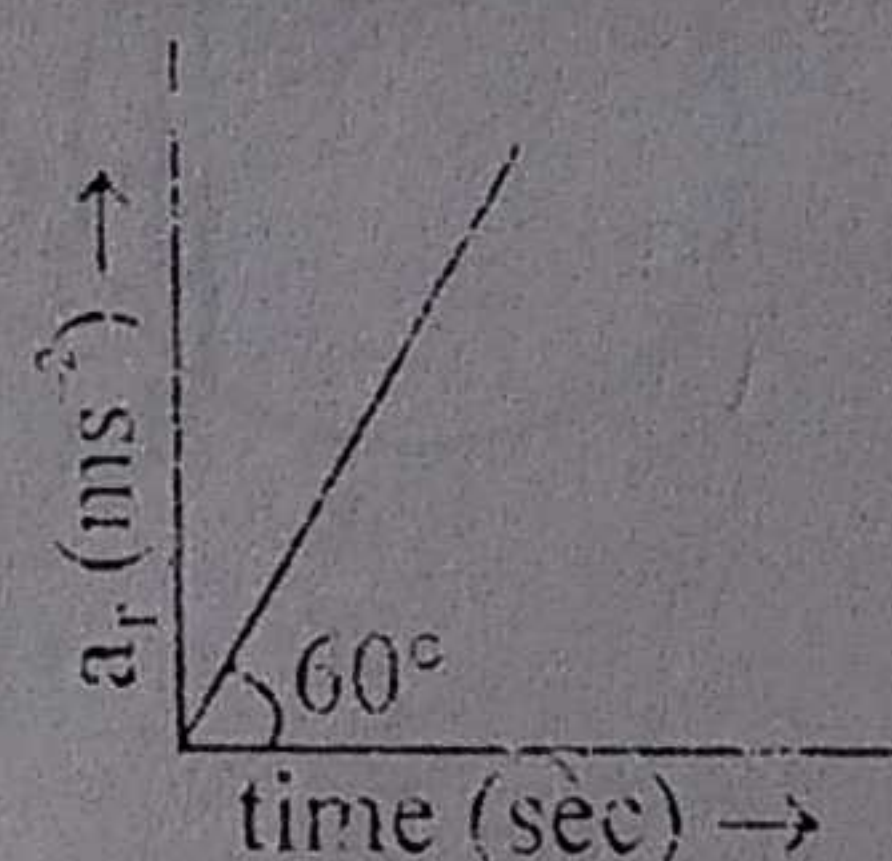
Q.5 It takes one minute for a passenger standing on an escalator to reach the top. If the escalator does not move it takes him 3 minute to walk up. How long will it take for the passenger to arrive at the top if he walks up the moving escalator?

- (A) 30 sec (B) 45 sec (C) 40 sec (D) 35 sec

Q.6 The co ordinates of a moving particle at a time t are give by $x = 5 \sin 10t$, $y = 5 \cos 10t$. The speed of the particle is :

- (A) 25 (B) 50 (C) 10 (D) None

Q.7 Tangential acceleration of a particle moving in a circle of radius 1 m varies with time t as (initial velocity of particle is zero). Time after which total acceleration of particle makes an angle of 30° with radial acceleration is



- (A) 4 sec (B) $4/3$ sec (C) $2^{2/3}$ sec (D) $\sqrt{2}$ sec

Q.8 A particle is projected from a horizontal plane (x-z plane) such that its velocity vector at time t is given by $\vec{v} = a\hat{i} + (b - ct)\hat{j}$. Its range on the horizontal plane is given by

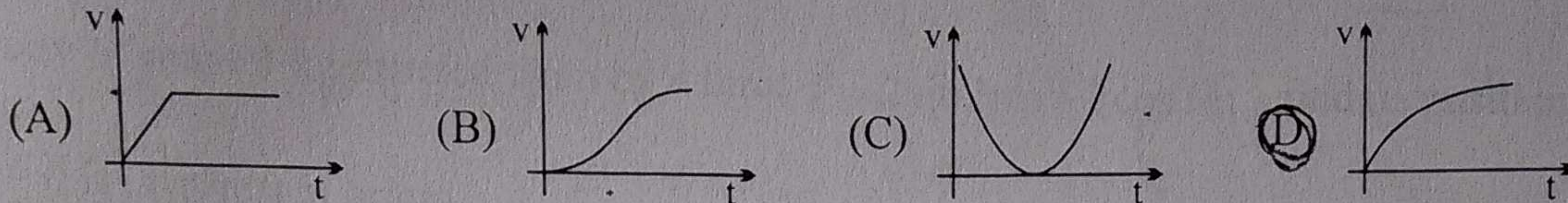
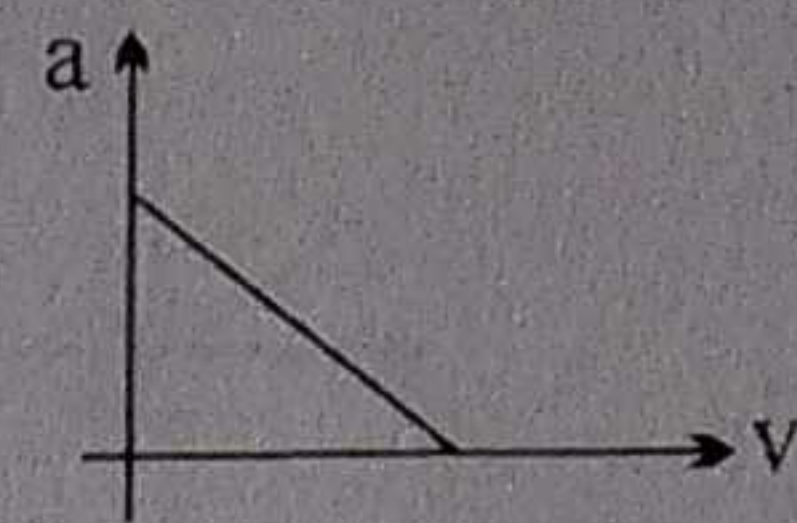
- (A) $\frac{ba}{c}$ (B) $\frac{2ba}{c}$ (C) $\frac{3ba}{c}$ (D) None

Q.9 A particle moves along a straight line in such a way that it's acceleration is increasing at the rate of 2 m/s^3 . It's initial acceleration and velocity were 0, the distance covered by it in $t = 3 \text{ second}$ is

- (A) 27m (B) 9m (C) 3m (D) 1m

- Q.10 A ball is thrown vertically down with velocity of 5m/s. With what velocity should another ball be thrown down after 2 seconds so that it can hit the 1st ball in 2 seconds
 (A) 40 m/s (B) 55 m/s (C) 15 m/s (D) 25 m/s

- Q.11 Acceleration versus velocity graph of a particle moving in a straight line starting from rest is as shown in figure. The corresponding velocity-time graph would be



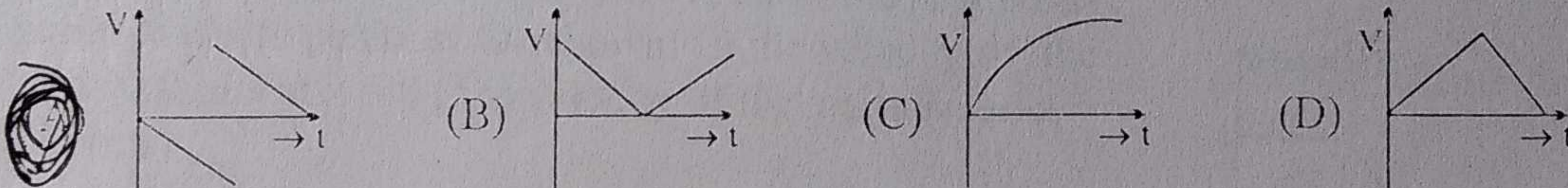
- Q.12 A particle is projected vertically upwards from a point A on the ground. It takes t_1 time to reach a point B but it still continues to move up. If it takes further t_2 time to reach the ground from point B then height of point B from the ground is

- (A) $\frac{1}{2}g(t_1 + t_2)^2$ (B) $g t_1 t_2$ (C) $\frac{1}{8}g(t_1 + t_2)^2$ (D) $\frac{1}{2} g t_1 t_2$

- Q.13 Balls are thrown vertically upward in such a way that the next ball is thrown when the previous one is at the maximum height. If the maximum height is 5m, the number of balls thrown per minute will be

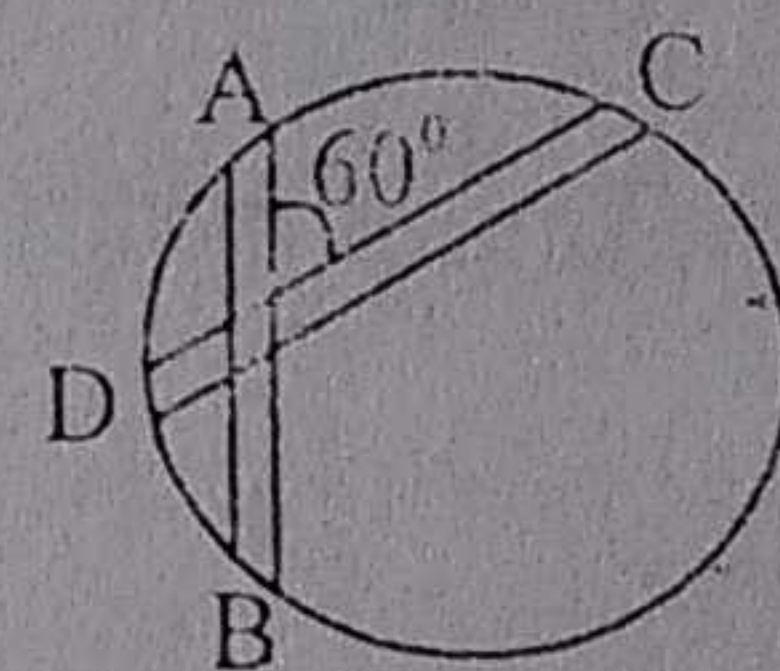
- (A) 40 (B) 50 (C) 60 (D) 120

- Q.14 The velocity-time graph of a body falling from rest under gravity and rebounding from a solid surface is represented by which of the following graphs?



- Q.15 A disc arranged in a vertical plane has two groves of same length directed along the vertical chord AB and CD as shown in the fig. The same particles slide

down along AB and CD. The ratio of the time t_{AB}/t_{CD} is



- (A) 1 : 2 (B) 1 : $\sqrt{2}$ (C) 2 : 1 (D) $\sqrt{2}$: 1

- Q.16 The magnitude of displacement of a particle moving in a circle of radius a with constant angular speed ω varies with time t as

- (A) $2 a \sin \omega t$ (B) $2a \sin \frac{\omega t}{2}$ (C) $2a \cos \omega t$ (D) $2a \cos \frac{\omega t}{2}$

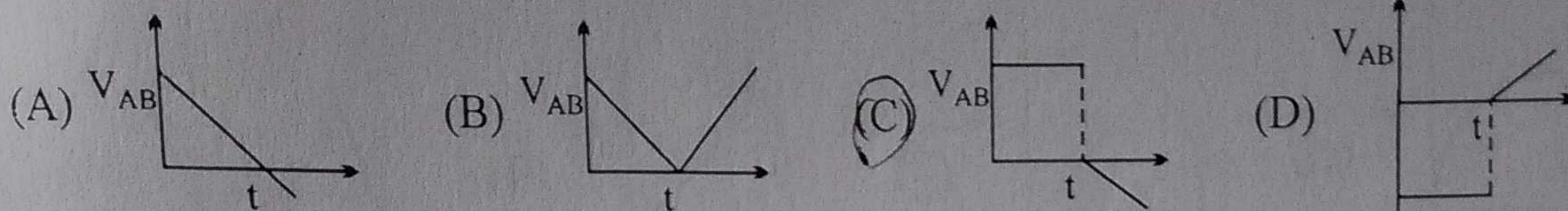
- Q.17 A body moves with velocity $v = \ln x$ m/s where x is its position. The net force acting on body is zero at

- (A) 0 m (B) $x = e^2$ m (C) $x = e$ m (D) $x = 1$ m

$$a = \frac{dv}{dt} = \frac{d}{dt} \ln x = \frac{1}{x} \frac{dx}{dt} = \frac{1}{x} v$$

nd th

- Q.18 A body A is thrown vertically upwards with such a velocity that it reaches a maximum height of h . Simultaneously another body B is dropped from height h . It strikes the ground and does not rebound. The velocity of A relative to B v/s time graph is best represented by : (upward direction is positive)



- Q.19 A body of mass 1 kg is acted upon by a force $\vec{F} = 2 \sin 3\pi t \hat{i} + 3 \cos 3\pi t \hat{j}$ find its position at $t = 1$ sec if at $t = 0$ it is at rest at origin.

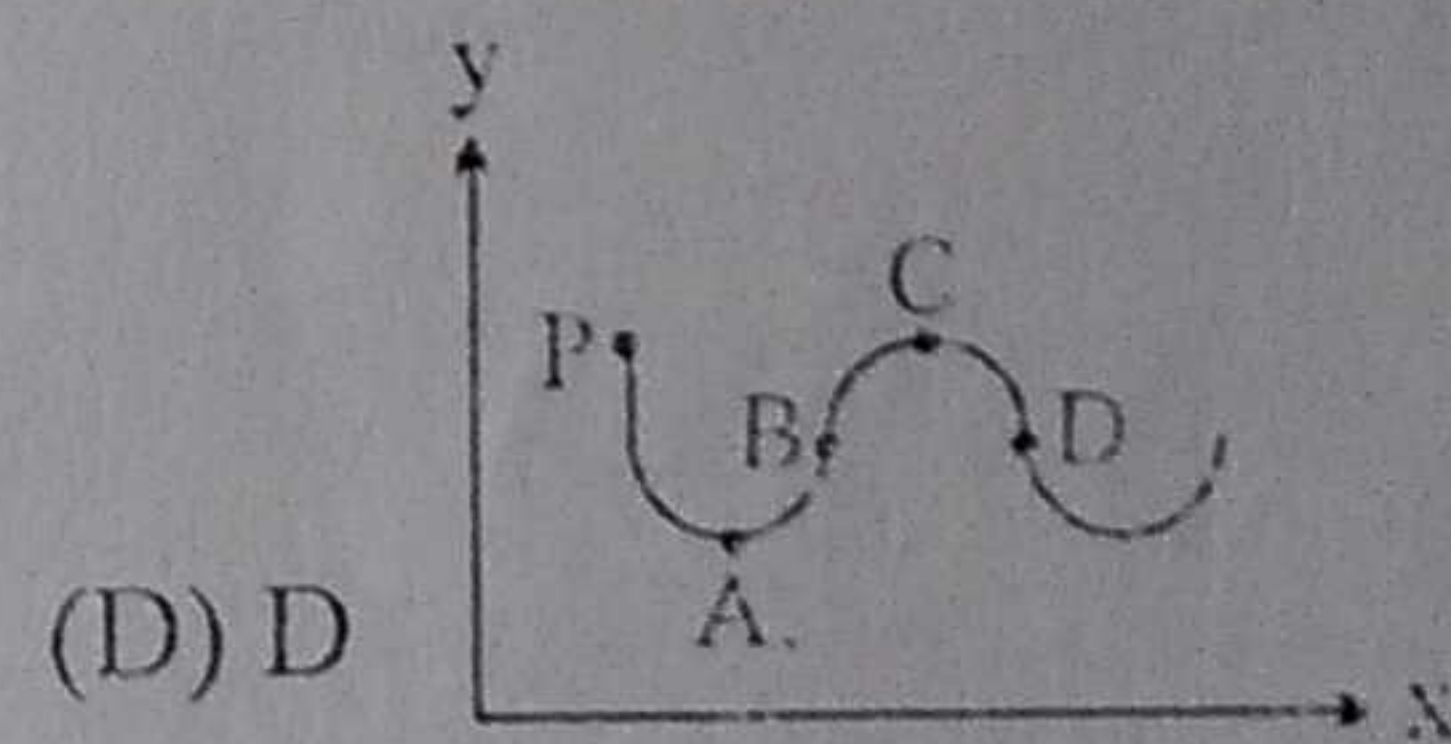
- (A) $\left(\frac{3}{3\pi^2}, \frac{2}{9\pi^2}\right)$ (B) $\left(\frac{2}{3\pi^2}, \frac{2}{3\pi^2}\right)$ (C) $\left(\frac{2}{3\pi}, \frac{2}{3\pi^2}\right)$ (D) none of these

- Q.20 A force $F = Be^{-Ct}$ acts on a particle whose mass is m and whose velocity is 0 at $t = 0$. Its terminal velocity is :

- (A) $\frac{C}{mB}$ (B) $\frac{B}{mC}$ (C) $\frac{BC}{m}$ (D) $-\frac{B}{mC}$

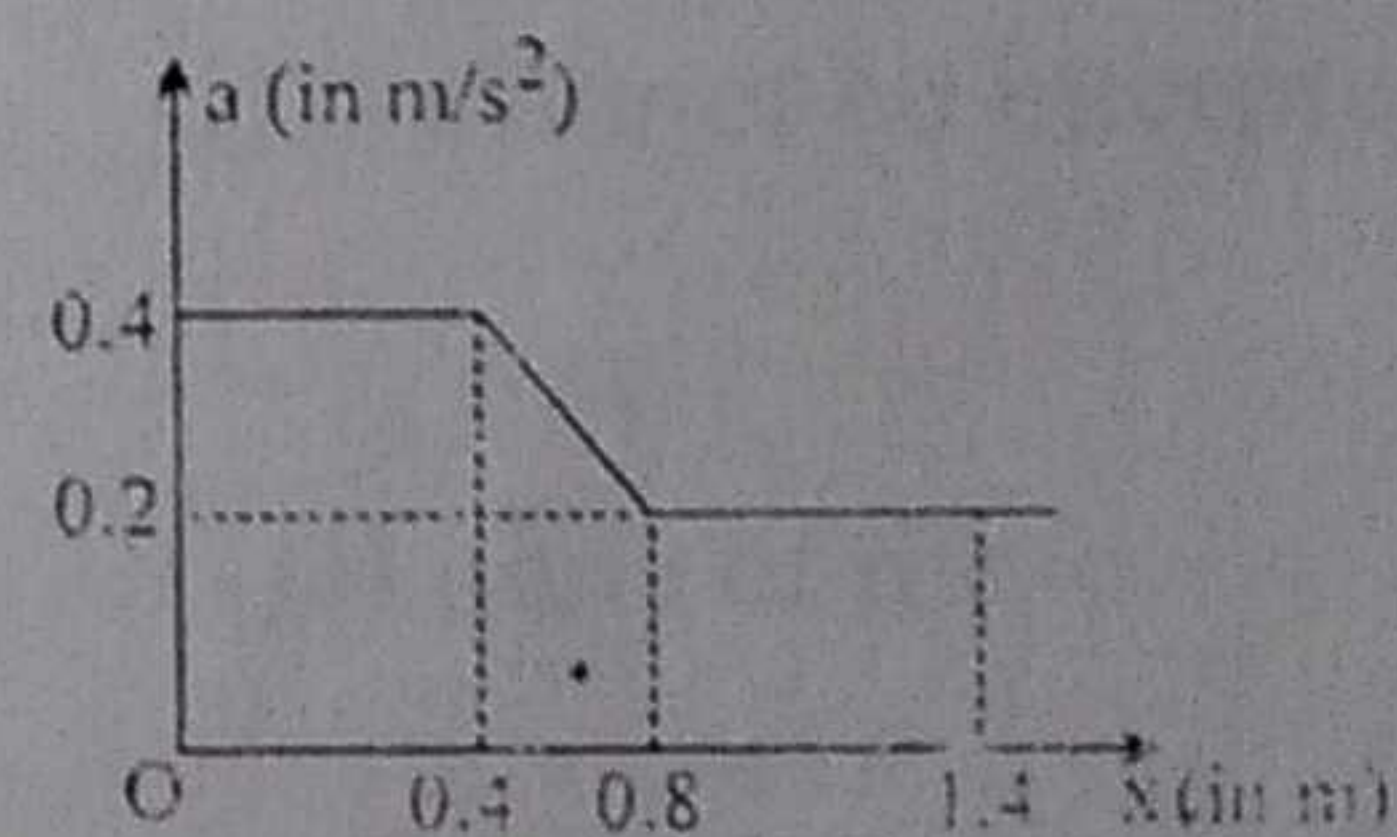
- Q.21 A man moves in x-y plane along the path shown. At what point is his average velocity vector in the same direction as his instantaneous velocity vector. The man starts from point P.

- (A) A (B) B (C) C (D) D



- Q.22 The acceleration of a particle which moves along the positive x-axis varies with its position as shown. If the velocity of the particle is 0.8 m/s at $x = 0$, the velocity of the particle at $x = 1.4$ is (in m/s)

- (A) 1.6 (B) 1.2 (C) 1.4 (D) none of these



- Q.23 A body is thrown up in a lift with a velocity u relative to the lift and the time of flight is found to be t . The acceleration with which the lift is moving up is

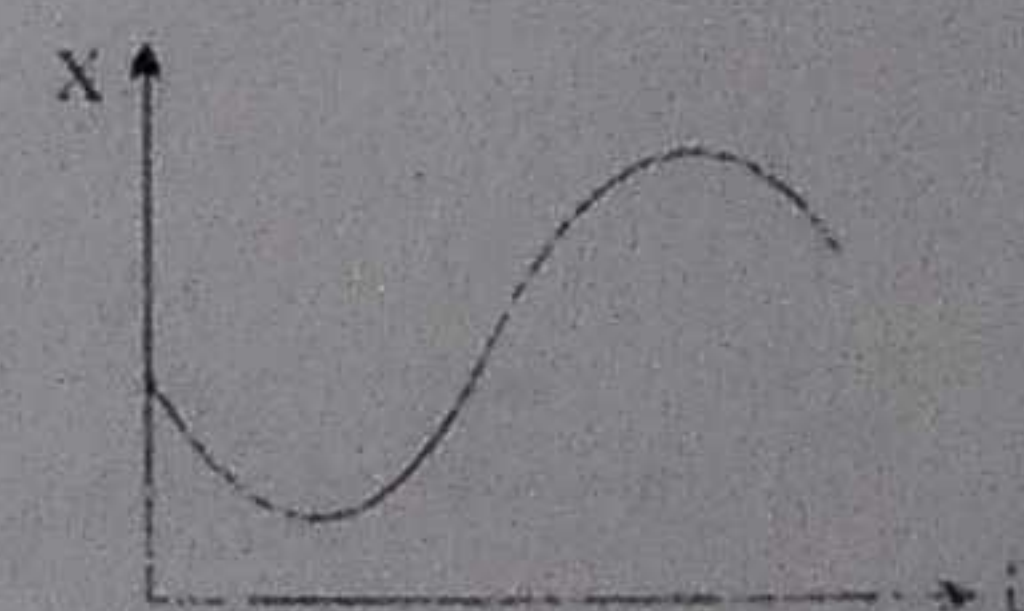
- (A) $\frac{u - gt}{t}$ (B) $\frac{2u - gt}{t}$ (C) $\frac{u + gt}{t}$ (D) $\frac{2u + gt}{t}$

- Q.24 A ball is thrown vertically downwards with velocity $\sqrt{2gh}$ from a height h . After colliding with the ground, it just reaches the starting point. Coefficient of restitution is

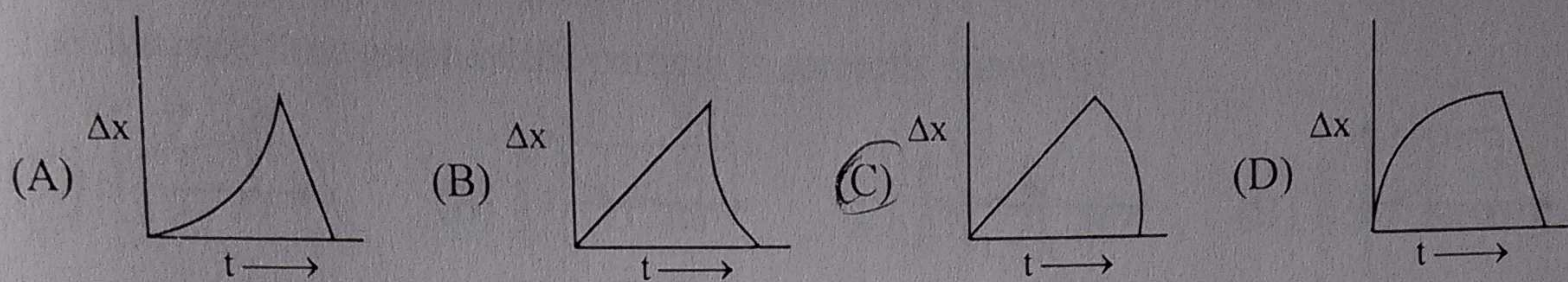
- (A) $1/\sqrt{2}$ (B) $1/2$ (C) 1 (D) $\sqrt{2}$

- Q.25 The graph of position x versus time t represents the motion of a particle. If b and c are both positive constants, which of the following expressions best describes the acceleration a of the particle?

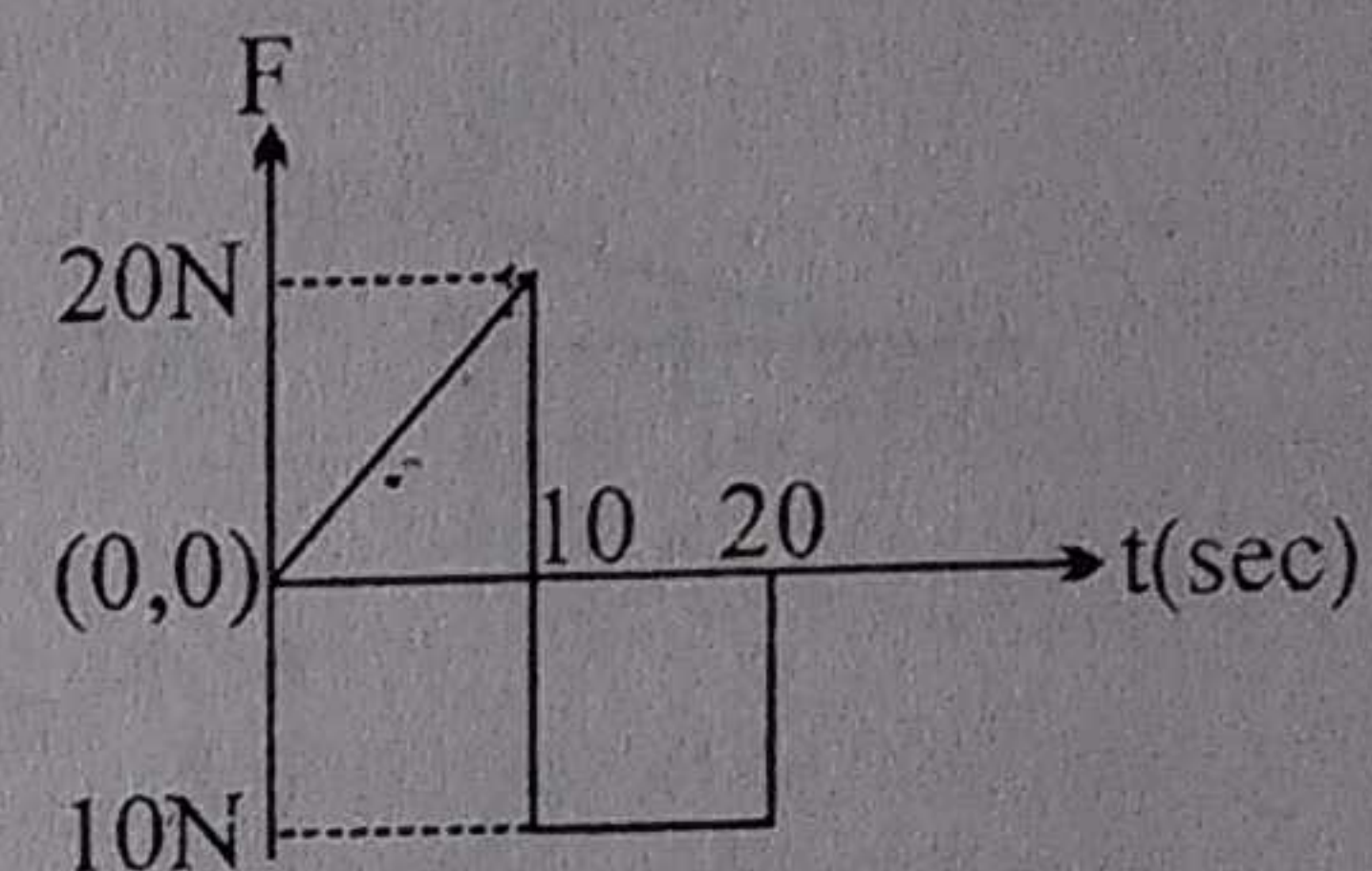
- (A) $a = b - ct$ (B) $a = +b$ (C) $a = -c$ (D) $a = b + ct$



Q.26 Two stones are thrown up vertically and simultaneously but with different speeds. Which graph correctly represents the time variation of their relative positions Δx . Assume that stones do not bounce after hitting ground.



Q.27 A particle of mass 1 kg is acted upon by a force 'F' which varies as shown in the figure. If initial velocity of the particle is 10 ms^{-1} , the maximum velocity attained by the particle during the period is
(A) 210 ms^{-1}
(B) 110 ms^{-1}
(C) 100 ms^{-1}
(D) 90 ms^{-1}



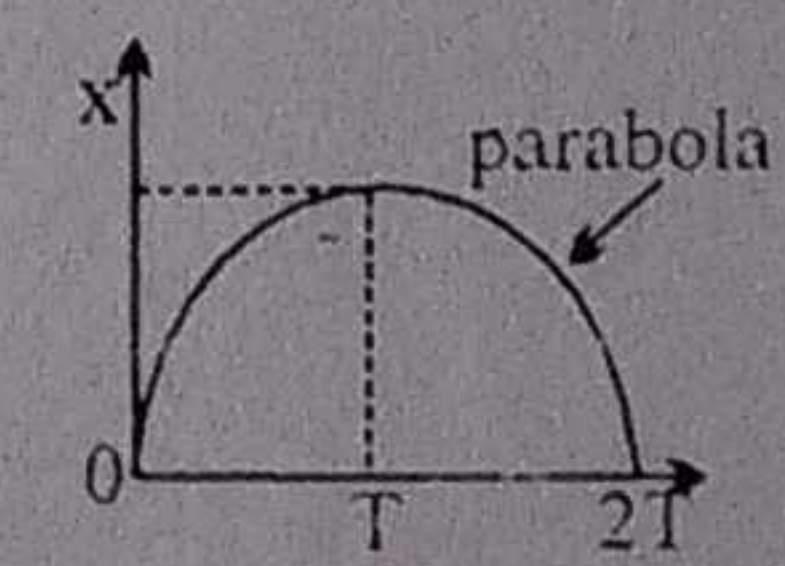
Question No. 28 to 32

Two balls A and B are thrown with same velocity u from the top of a tower. Ball A is thrown vertically upwards and the ball B is thrown vertically downwards.

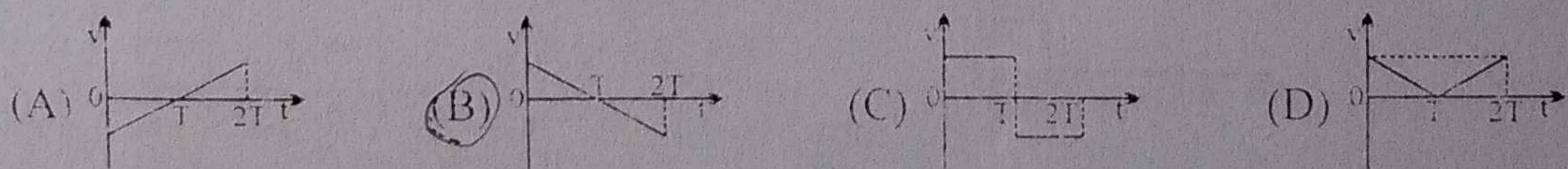
- Q.28 Choose the correct statement
(A) Ball B reaches the ground with greater velocity
(B) Ball A reaches the ground with greater velocity
(C) Both the balls reach the ground with same velocity
(D) Cannot be interpreted
- Q.29 If t_A and t_B are the respective times taken by the balls A and B respectively to reach the ground, then identify the correct statement
(A) $t_A > t_B$ (B) $t_A = t_B$ (C) $t_A < t_B$ (D) Cannot be interpreted
- Q.30 If $t_A = 6 \text{ s}$ and $t_B = 2 \text{ s}$, then the height of the tower is
(A) 80 m (B) 60 m (C) 45 m (D) none of these
- Q.31 The velocity u of each ball is
(A) 10 ms^{-1} (B) 15 ms^{-1} (C) 20 ms^{-1} (D) none of these
- Q.32 If a ball C is thrown with the same velocity but in the horizontal direction from the top of the tower, then it will reach the ground in time t_c equal to
(A) 4 s (B) 3.46 s (C) 4.2 s (D) none of these

Question No. 33 to 36

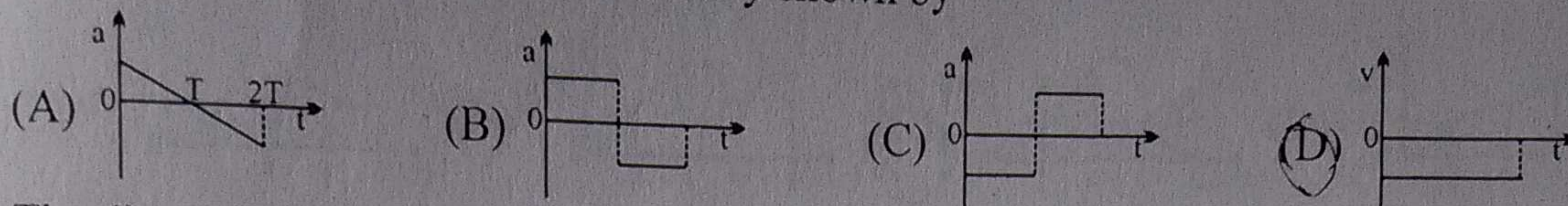
The x-t graph of a particle moving along a straight line is shown in figure



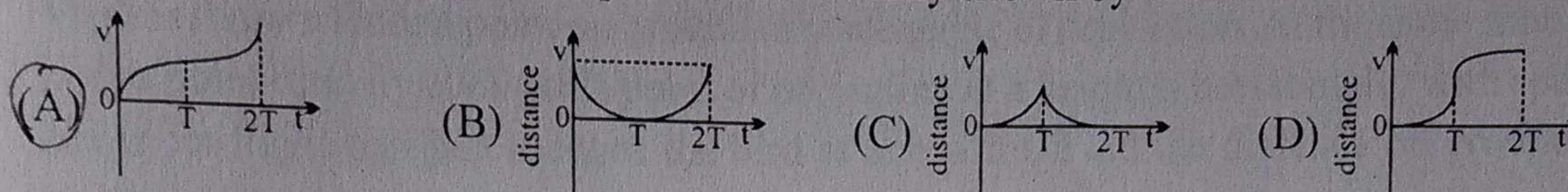
Q.33 The v-t graph of the particle is correctly shown by



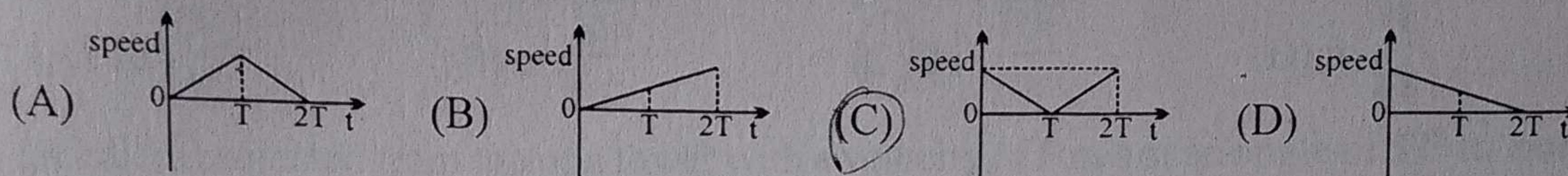
Q.34 The a-t graph of the particle is correctly shown by



Q.35 The distance-time graph of the particle is correctly shown by

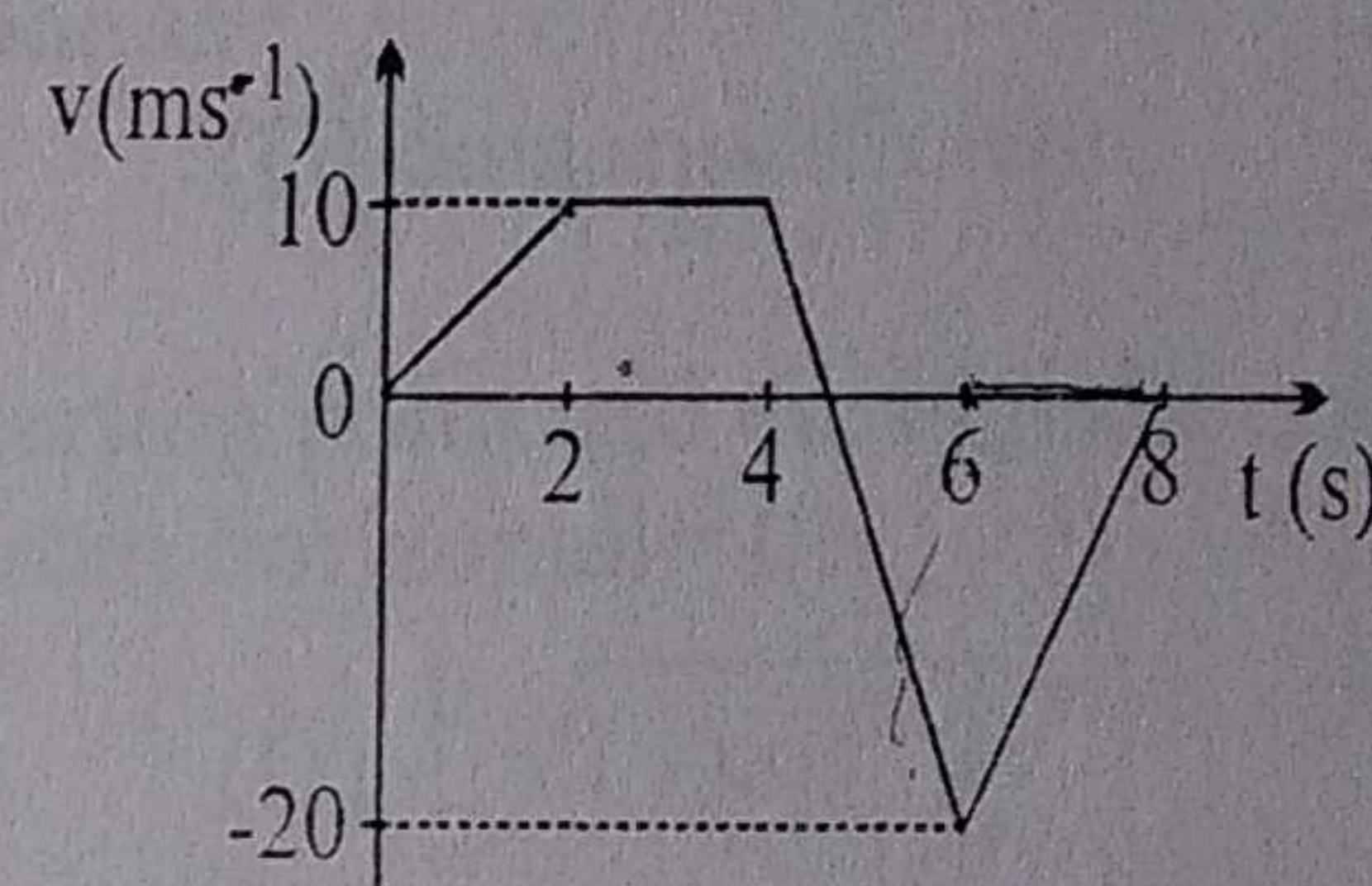


Q.36 The speed-time graph of the particle is correctly shown by



Question No. 37 to 43

The figure shows a velocity-time graph of a particle moving along a straight line



Q.37 Choose the incorrect statement. The particle comes to rest at
(A) $t = 0$ s (B) $t = 5$ s
(C) $t = 8$ s (D) none of these

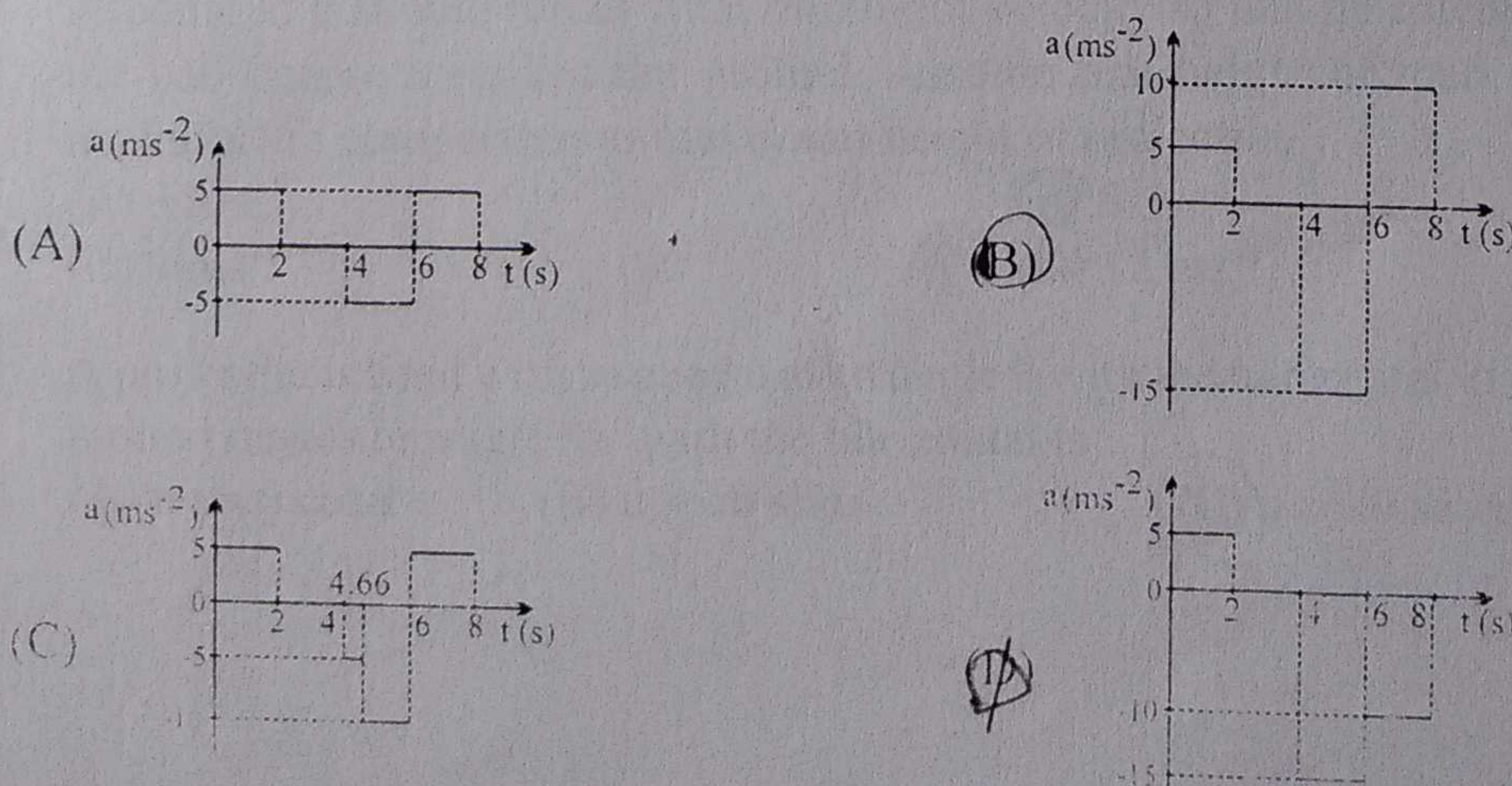
Q.38 Identify the region in which the rate of change of velocity $\left| \frac{\Delta \vec{v}}{\Delta t} \right|$ of the particle is maximum
(A) 0 to 2s (B) 2 to 4s (C) 4 to 6 s (D) 6 to 8 s

Q.39 If the particle starts from the position $x_0 = -15$ m, then its position at $t = 2$ s will be
(A) -5 m (B) 5 m (C) 10 m (D) 15 m

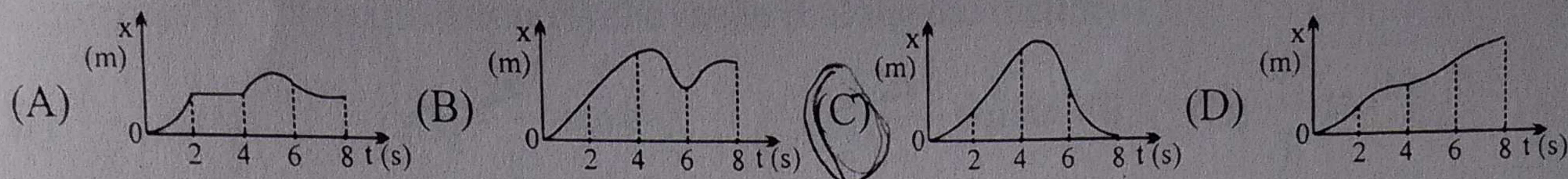
Q.40 The maximum displacement of the particle is
(A) 33.3 m (B) 23.3 m (C) 18.3 m (D) zero

Q.41 The total distance travelled by the particle is
(A) 66.6 m (B) 51.6 m (C) zero (D) 36.6 m

Q.42 The correct acceleration-time graph of the particle is shown as



43 The correct displacement-time graph of the particle is shown as



44 A ball is thrown from a point on ground at some angle of projection. At the same time a bird starts from a point directly above this point of projection at a height h horizontally with speed u . Given that in its flight ball just touches the bird at one point. Find the distance on ground where ball strikes

- (A) $2u\sqrt{\frac{h}{g}}$ (B) $u\sqrt{\frac{2h}{g}}$ (C) $2u\sqrt{\frac{2h}{g}}$ (D) $u\sqrt{\frac{h}{g}}$

Q.45 A ball is projected from top of a tower with a velocity of 5 m/s at an angle of 53° to horizontal. Its speed when it is at a height of 0.45 m from the point of projection is :

- (A) 2 m/s (B) 3 m/s (C) 4 m/s (D) data insufficient.

Q.46 Average velocity of a particle in projectile motion between its starting point and the highest point of its trajectory is : (projection speed = u , angle of projection from horizontal = θ)

- (A) $u \cos \theta$ (B) $\frac{u}{2} \sqrt{1 + 3 \cos^2 \theta}$ (C) $\frac{u}{2} \sqrt{2 + \cos^2 \theta}$ (D) $\frac{u}{2} \sqrt{1 + \cos^2 \theta}$

Q.47 Find time of flight of projectile thrown horizontally with speed 50 ms^{-1} from a long inclined plane which makes an angle of $\theta = 45^\circ$ from horizontal.

- (A) $\sqrt{2} \text{ sec}$ (B) $2\sqrt{2} \text{ sec}$ (C) 2 sec (D) none

Q.48 Particle is dropped from the height of 20 m from horizontal ground. There is wind blowing due to which horizontal acceleration of the particle becomes 6 ms^{-2} . Find the horizontal displacement of the particle till it reaches ground.

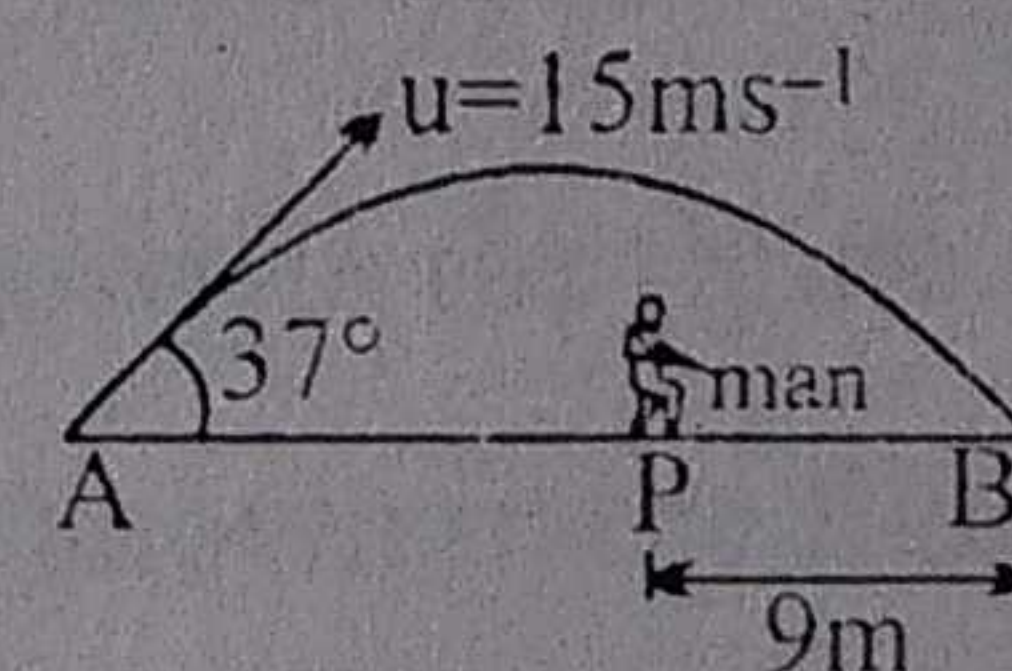
- (A) 6 m (B) 10 m (C) 12 m (D) 24 m

Q.49 A ball is dropped from height 5 m . The time after which ball stops rebounding if coefficient of restitution between ball and ground $e = 1/2$, is

- (A) 1 sec (B) 2 sec (C) 3 sec (D) infinite

Q.50 A ball is hit by a batsman at an angle of 37° as shown in figure. The man standing at P should run at what minimum velocity so that he catches the ball before it strikes the ground. Assume that height of man is negligible in comparison to maximum height of projectile.

- (A) 3 ms^{-1} (B) 5 ms^{-1} (C) 9 ms^{-1} (D) 12 ms^{-1}



Q.51 A projectile is fired with a speed u at an angle θ with the horizontal. Its speed when its direction of motion makes an angle ' α ' with the horizontal is

- (A) $u \sec \theta \cos \alpha$ (B) $u \sec \theta \sin \alpha$ (C) $u \cos \theta \sec \alpha$ (D) $u \sin \theta \sec \alpha$

Q.52 A projectile is fired with a velocity at right angle to the slope which is inclined at an angle θ with the horizontal. The expression for the range R along the incline is

- (A) $\frac{2v^2}{g} \sec \theta$ (B) $\frac{2v^2}{g} \tan \theta$ (C) $\frac{2v^2}{g} \tan \theta \sec \theta$ (D) $\frac{v^2}{g} \tan^2 \theta$

Q.53 A particle is projected vertically upwards from O with velocity v and a second particle is projected at the same instant from P (at a height h above O) with velocity v at an angle of projection θ . The time when the distance between them is minimum is

- (A) $\frac{h}{2v \sin \theta}$ (B) $\frac{h}{2v \cos \theta}$ (C) h/v (D) $h/2v$

Q.54 A ball is projected from ground with a velocity V at an angle θ to the vertical. On its path it makes an elastic collision with a vertical wall and returns to ground. The total time of flight of the ball is

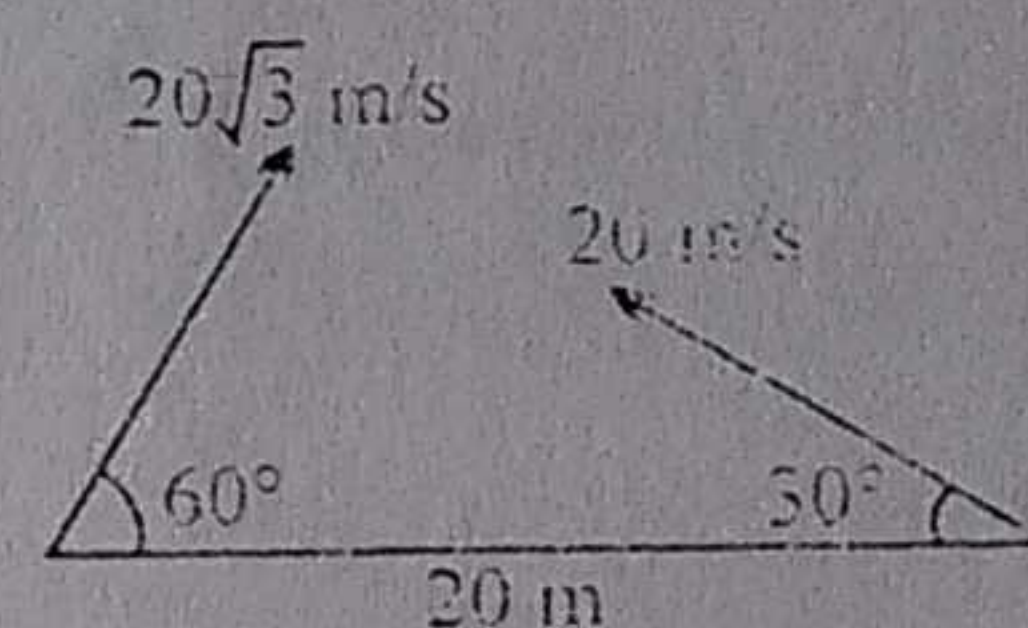
- (A) $\frac{2v \sin \theta}{g}$ (B) $\frac{2v \cos \theta}{g}$ (C) $\frac{v \sin 2\theta}{g}$ (D) $\frac{v \cos \theta}{g}$

Q.55 Two particles are moving along two long straight lines, in the same plane, with the same speed = 20 cm/s. The angle between the two lines is 60° , and their intersection point is O . At a certain moment, the two particles are located at distances 3m and 4m from O , and are moving towards O . Subsequently, the shortest distance between them will be

- (A) 50 cm (B) $40\sqrt{2}$ cm (C) $50\sqrt{2}$ cm (D) $50\sqrt{3}$ cm

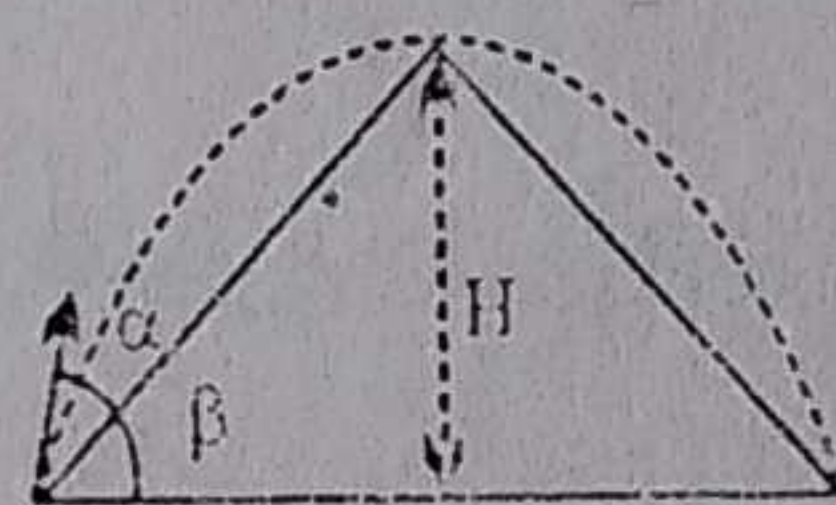
Q.56 In the figure shown, the two projectiles are fired simultaneously. The minimum distance between them during their flight is

- (A) 20 m (B) $10\sqrt{3}$ m (C) 10 m (D) None



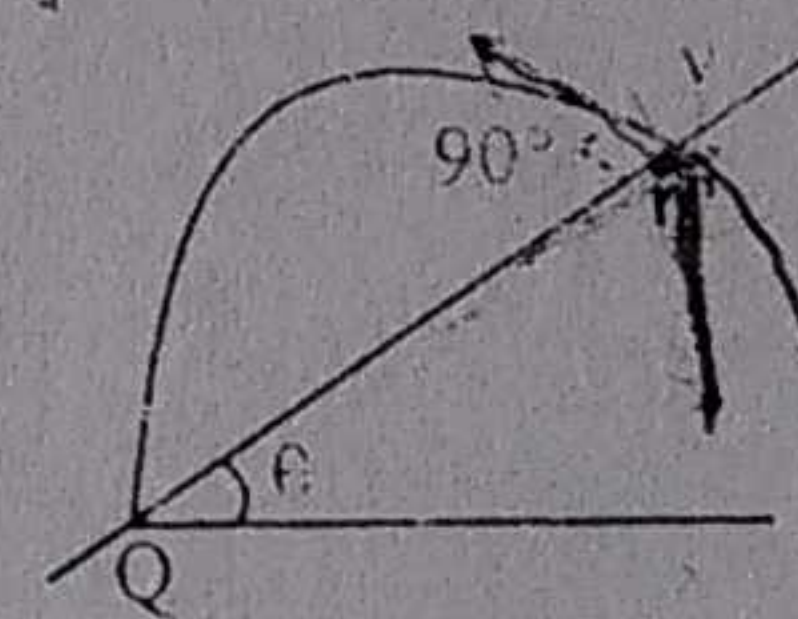
Q.57 A shell fired from the base of a mountain just clears it. If α is the angle of projection then the angular elevation of the summit β is

- (A) $\frac{1}{2} \alpha$ (B) $\tan^{-1}(1/2)$
(C) $\tan^{-1}(1/2 \tan \alpha)$ (D) $\tan^{-1}(2 \tan \alpha)$



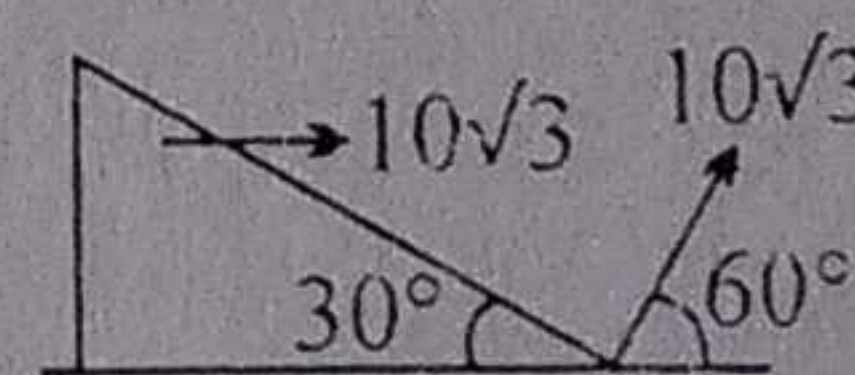
Q.58 If time taken by the projectile to reach Q is T , then $PQ =$

- (A) $Tv \sin \theta$ (B) $Tv \cos \theta$ (C) $Tv \sec \theta$ (D) $Tv \tan \theta$



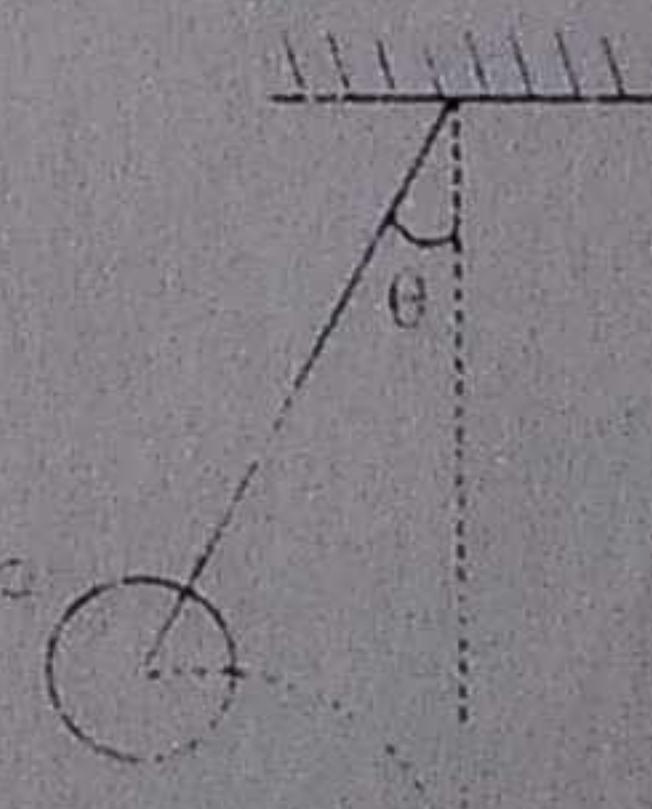
Q.59 A particle is projected at angle 60° with speed $10\sqrt{3}$, from the point 'A' as shown in the figure. At the same time the wedge is made to move with speed $10\sqrt{3}$ towards right as shown in the figure. Then the time after which particle will strike with wedge is

- (A) 2 sec (B) $2\sqrt{3}$ sec (C) $\frac{4}{\sqrt{3}}$ sec (D) None



Q.60 A ball is held in the position shown with string of length 1 m just taut & then projected horizontally with a velocity of 3 m/s. If the string becomes taut again when it is vertical, angle θ is given by

- (A) 53° (B) 30° (C) 45° (D) 37°



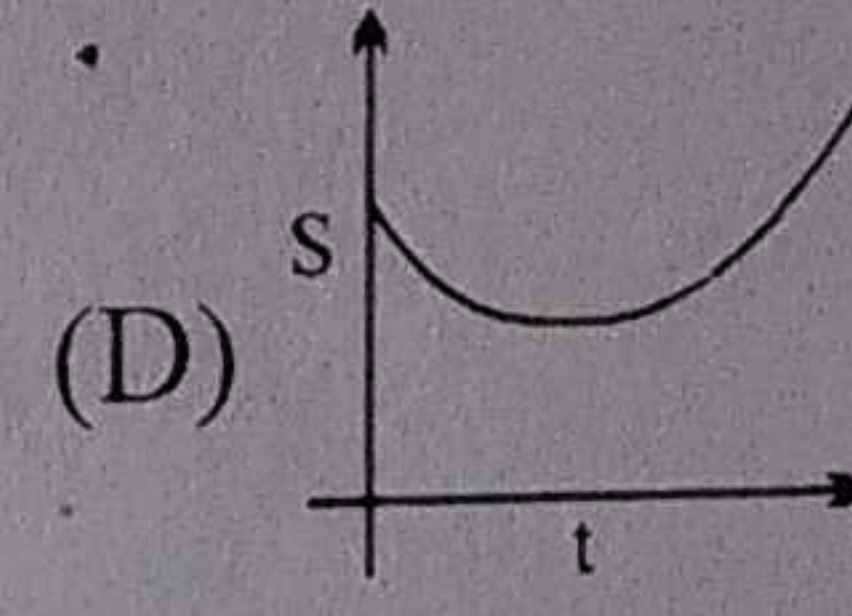
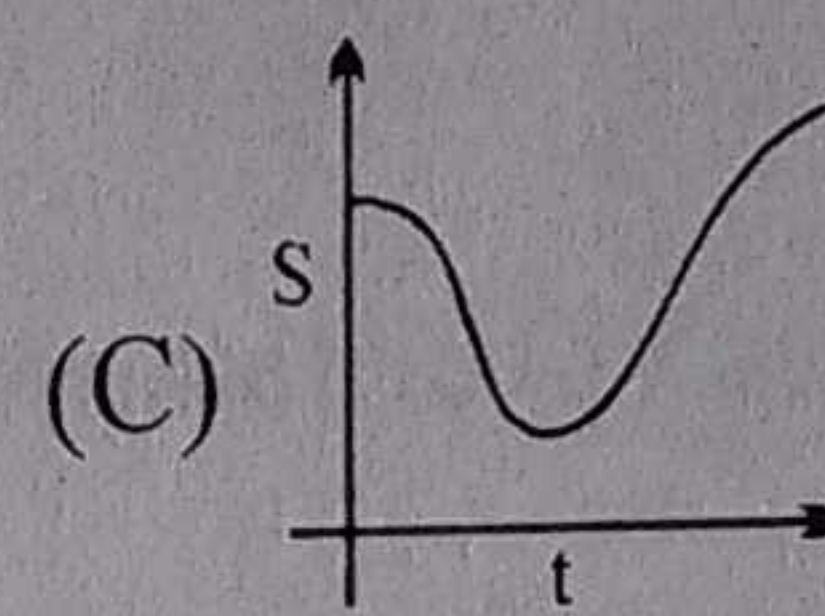
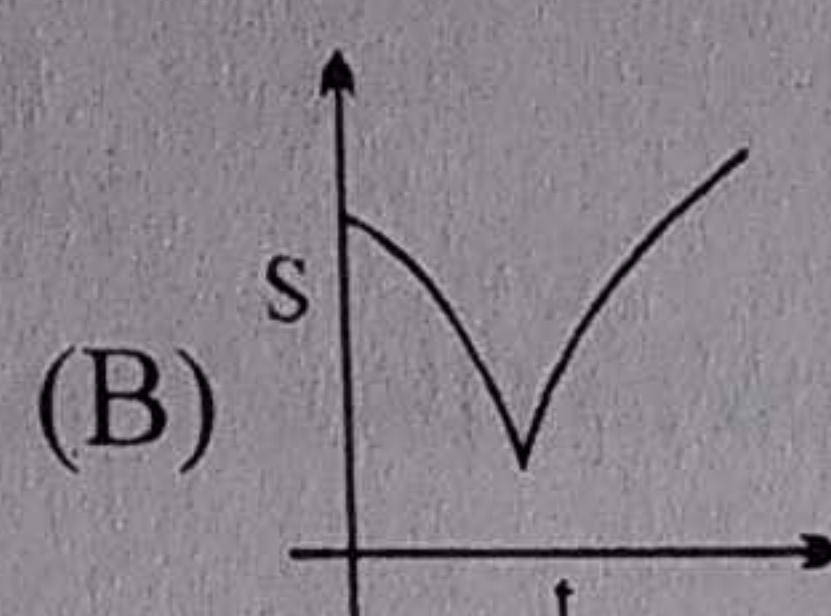
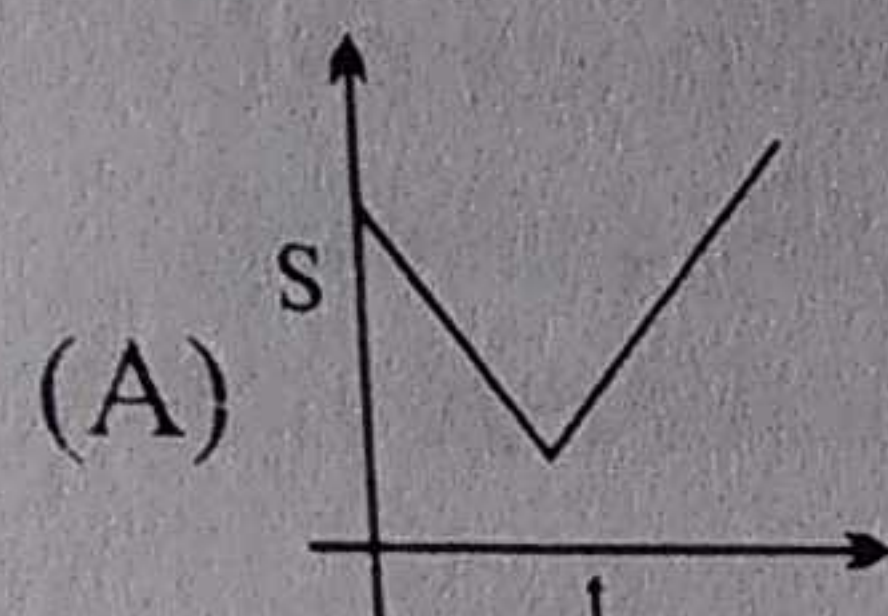
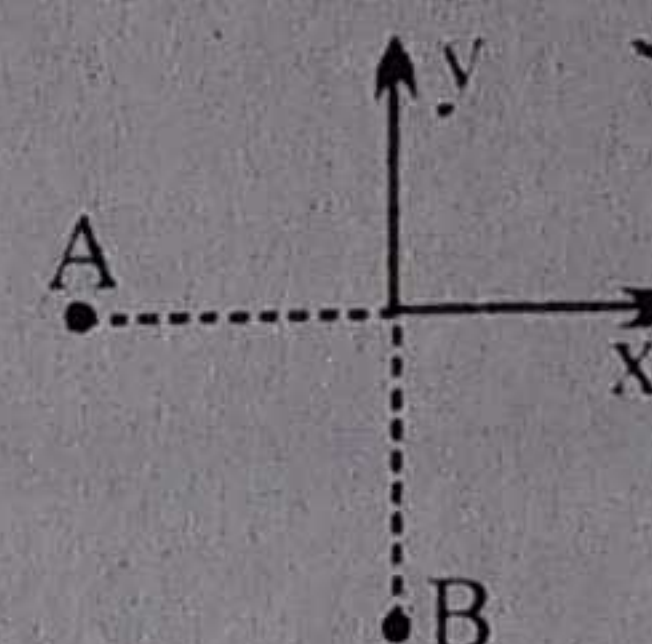
Q.61 Two projectiles A and B are thrown with the same speed such that A makes angle θ with the horizontal and B makes angle θ with the vertical, then

(A) Both must have same time of flight (B) Both must achieve same maximum height

(C) A must have more horizontal range than B

(D) Both may have same time of flight *time of flight Range*

Q.62 Particles A and B are moving with constant velocities along x and y axis respectively, the graph of separation between them with time is



Q.63 An aeroplane flying at a constant velocity releases a bomb. As the bomb drops down from the aeroplane,

(A) it will always be vertically below the aeroplane

(B) it will always be vertically below the aeroplane only if the aeroplane is flying horizontally

(C) it will always be vertically below the aeroplane only if the aeroplane is flying at an angle of 45° to the horizontal

(D) it will gradually fall behind the aeroplane if the aeroplane is flying horizontally

Q.64 Two particles are projected simultaneously in the same vertical plane, from the same point on ground, but with same speeds but at different angles ($< 90^\circ$) to the horizontal. The path followed by one, as seen by the other, is

(A) a vertical straight line

(B) a straight line making a constant angle with the horizontal

(C) a parabola

(D) a hyperbola

Q.65 Suppose a player hits several baseballs. Which baseball will be in the air for the longest time?

(A) The one with the farthest range.

(B) The one which reaches maximum height.

(C) The one with the greatest initial velocity.

(D) The one leaving the bat at 45° with respect to the ground.

Question No. 66 to 69

Two projectiles are thrown simultaneously in the same plane from the same point. If their velocities are v_1 and v_2 at angles θ_1 and θ_2 respectively from the horizontal, then answer the following questions

Q.66 The trajectory of particle 1 with respect to particle 2 will be

(A) a parabola

(B) a straight line

(C) a vertical straight line

(D) a horizontal straight line

Q.67 If $v_1 \cos \theta_1 = v_2 \cos \theta_2$, then choose the incorrect statement

(A) one particle will remain exactly below or above the other particle

(B) the trajectory of one with respect to other will be a vertical straight line

(C) both will have the same range

(D) none of these

- Q.68 If $v_1 \sin \theta_1 = v_2 \sin \theta_2$, then choose the incorrect statement
 (A) the time of flight of both the particles will be same
 (B) the maximum height attained by the particles will be same
 (C) the trajectory of one with respect to another will be a horizontal straight line
 (D) none of these

- Q.69 If $v_1 = v_2$ and $\theta_1 > \theta_2$, then choose the incorrect statement
 (A) Particle 2 moves under the particle 1
 (B) The slope of the trajectory of particle 2 with respect to 1 is always positive
 (C) Both the particle will have the same range if $\theta_1 > 45^\circ$ and $\theta_2 < 45^\circ$ and $\theta_1 + \theta_2 = 90^\circ$
 (D) none of these

Question No. 70 to 75

A projectile crosses two walls of equal height H symmetrically as shown

- Q.70 The time of flight T is given by
 (A) 8 s (B) 9 s (C) 7 s (D) 10 s
- Q.71 The height of each wall is
 (A) 240 m (B) 120 m (C) 60 m (D) 30 m
- Q.72 The maximum height of the projectile is
 (A) 120 m (B) 80 m (C) 160 m (D) cannot be obtained
- Q.73 If the horizontal distance between the two walls is $d = 120$ m, then the range of the projectile is
 (A) 240 m (B) 160 m (C) 300 m (D) cannot be obtained
- Q.74 The angle of projection of the projectile is
 (A) $\tan^{-1}(3/4)$ (B) $\tan^{-1}(4/3)$ (C) $\tan^{-1}(4/5)$ (D) $\tan^{-1}(3/5)$
- Q.75 The velocity of projection is
 (A) 30 ms^{-1} (B) 40 ms^{-1} (C) 50 ms^{-1} (D) none of these

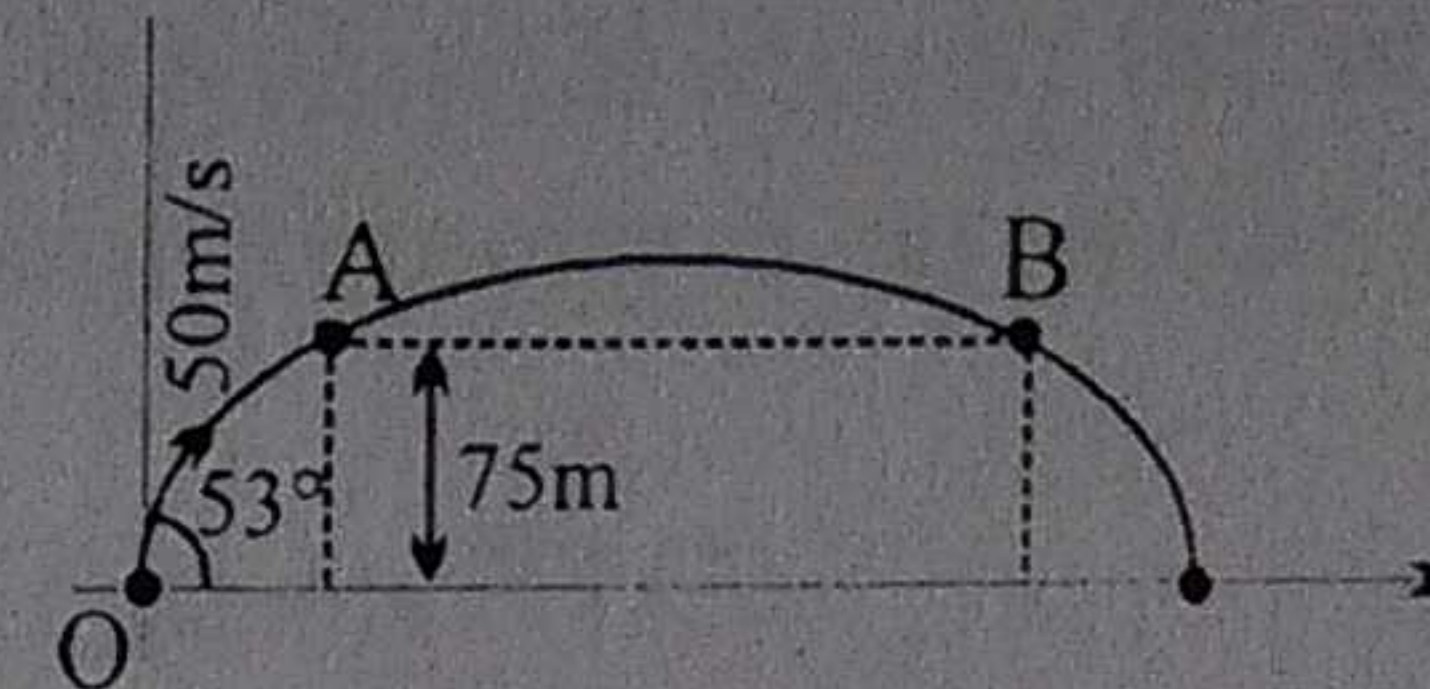
Question No. 76 to 78

A projectile is thrown with a velocity of 50 ms^{-1} at an angle of 53° with the horizontal

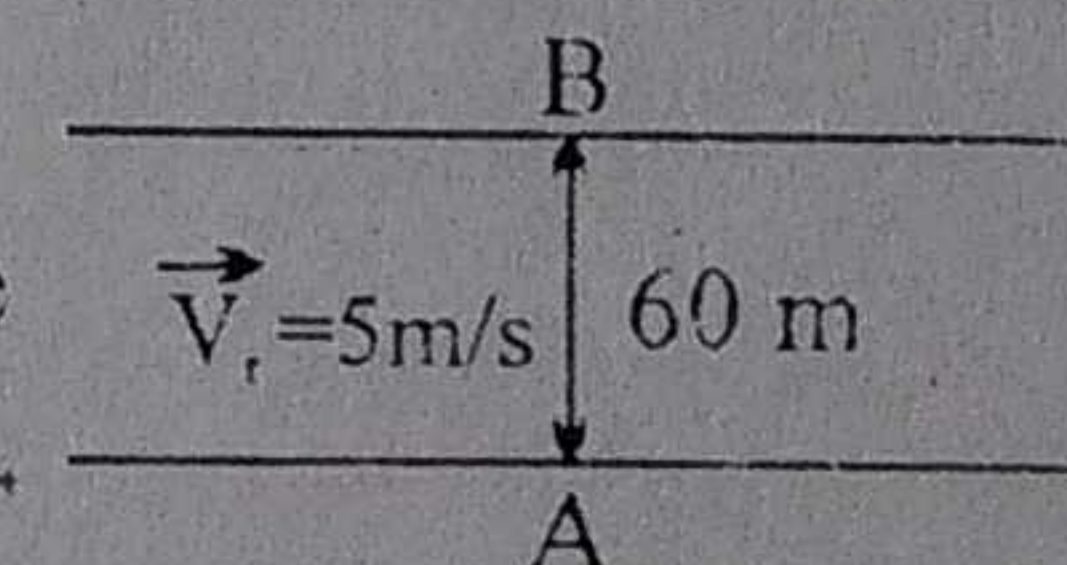
- Q.76 Choose the incorrect statement
 (A) It travels vertically with a velocity of 40 ms^{-1}
 (B) It travels horizontally with a velocity of 30 ms^{-1}
 (C) The minimum velocity of the projectile is 30 ms^{-1}
 (D) None of these
- Q.77 Determine the instants at which the projectile is at the same height
 (A) $t = 1$ s and $t = 7$ s (B) $t = 3$ s and $t = 5$ s (C) $t = 2$ s and $t = 6$ s (D) all the above
- Q.78 The equation of the trajectory is given by
 (A) $180y = 240x - x^2$ (B) $180y = x^2 - 240x$
 (C) $180y = 135x - x^2$ (D) $180y = x^2 - 135x$

Question No. 79 & 80

At $t = 0$ a projectile is fired from a point O (taken as origin) on the ground with a speed of 50 m/s at an angle of 53° with the horizontal. It just passes two points A & B each at height 75 m above horizontal as shown.



- Q.79 The horizontal separation between the points A and B is
 (A) 30 m (B) 60 m (C) 90 m (D) None
- Q.80 The distance (in metres) of the particle from origin at $t = 2$ sec.
 (A) $60\sqrt{2}$ (B) 100 (C) 60 (D) 120
- Q.81 A particle starts moving rectilinearly at time $t = 0$ such that its velocity ' v ' changes with time ' t ' according to the equation $v = t^2 - t$ where t is in seconds and v is in m/s. The time interval for which the particle retards is
 (A) $t < 1/2$ (B) $1/2 < t < 1$ (C) $t > 1$ (D) $t < 1/2$ and $t > 1$
- Q.82 A swimmer swims in still water at a speed = 5 km/hr. He enters a 200 m wide river, having river flow speed = 4 km/hr at point A and proceeds to swim at an angle of 127° ($\sin 37^\circ = 0.6$) with the river flow direction. Another point B is located directly across A on the other side. The swimmer lands on the other bank at a point C, from which he walks the distance CB with a speed = 3 km/hr. The total time in which he reaches from A to B is
 (A) 5 minutes (B) 4 minutes (C) 3 minutes (D) None
- Q.83 A boat having a speed of 5 km/hr. in still water, crosses a river of width 1 km along the shortest possible path in 15 minutes. The speed of the river in Km/hr.
 (A) 1 (B) 3 (C) 4 (D) $\sqrt{41}$
- Q.84 Two particles start simultaneously from the same point and move along two straight lines, one with uniform velocity v and other with a uniform acceleration a . If α is the angle between the lines of motion of two particles then the least value of relative velocity will be at time given by
 (A) $(v/a) \sin \alpha$ (B) $(v/a) \cos \alpha$ (C) $(v/a) \tan \alpha$ (D) $(v/a) \cot \alpha$
- Q.85 A flag is mounted on a car moving due North with velocity of 20 km/hr. Strong winds are blowing due East with velocity of 20 km/hr. The flag will point in direction
 (A) East (B) North - East (C) South - East (D) South - West
- Q.86 A man is crossing a river flowing with velocity of 5 m/s. He reaches a point directly across at a distance of 60 m in 5 sec. His velocity in still water should be
 (A) 12 m/s (B) 13 m/s (C) 5 m/s (D) 10 m/s
- Q.87 A man swimming down stream overcome a float at a point M. After travelling distance D he turned back and passed the float at a distance of $D/2$ from the point M, then the ratio of speed of swimmer with respect to still water to the speed of the river will be
 (A) 2 (B) 3 (C) 4 (D) 2.5



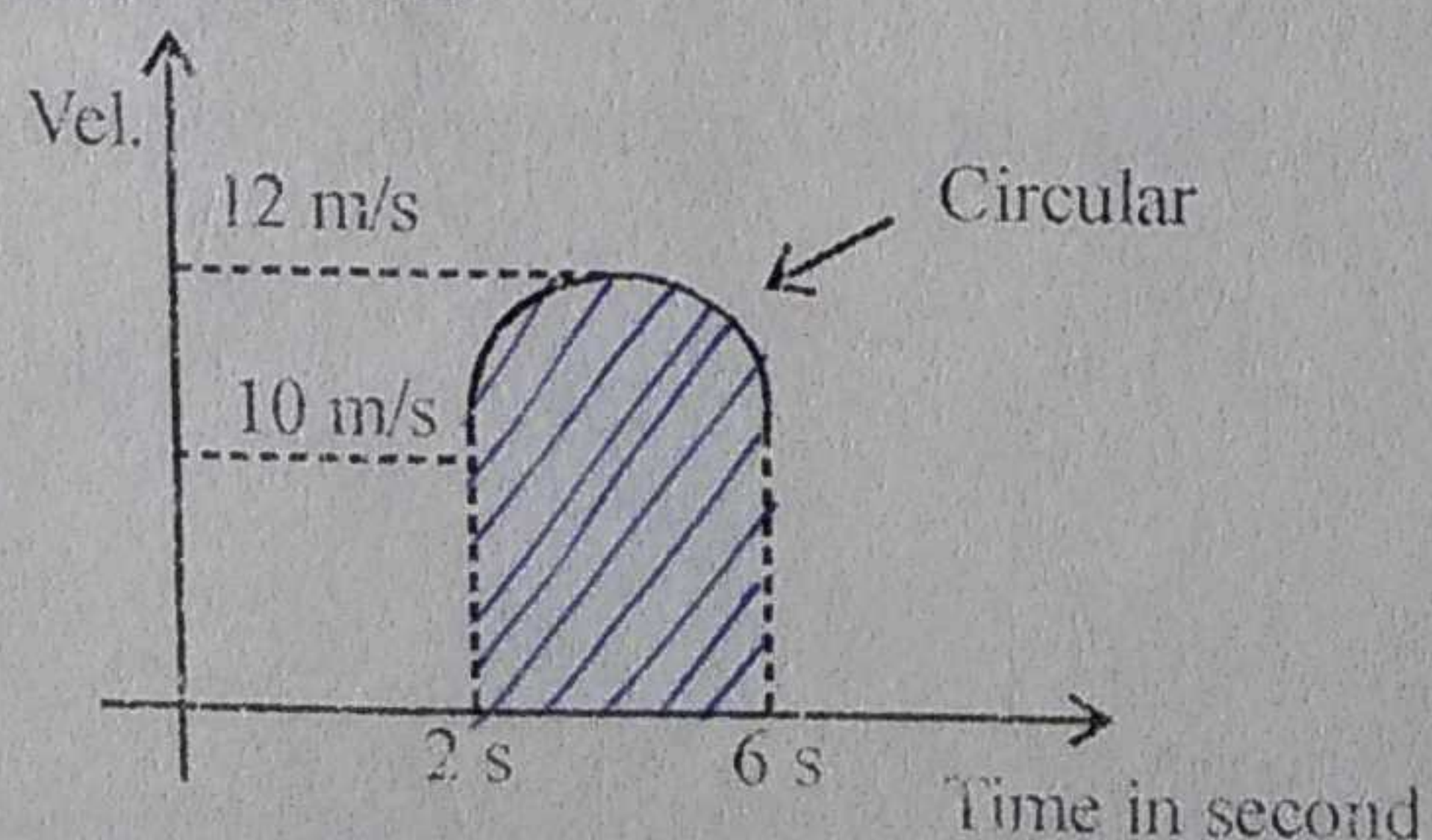
- Q.88 A glass wind screen whose inclination with the vertical can be changed is mounted on a car. The car moves horizontally with a speed of 2 m/s. At what angle α with the vertical should the wind screen be placed so that the rain drops falling vertically downwards with velocity 6 m/s strike the wind screen perpendicularly.
 (A) $\tan^{-1}(3)$ (B) $\tan^{-1}(1/3)$ (C) $\cos^{-1}(3)$ (D) $\sin^{-1}(1/3)$
- Q.89 Wind is blowing in the north direction at speed of 2 m/s which causes the rain to fall at some angle with the vertical. With what velocity should a cyclist drive so that the rain appears vertical to him :
 (A) 2 m/s south (B) 2 m/s north (C) 4 m/s west (D) 4 m/s south
- Q.90 Three particles, located initially on the vertices of an equilateral triangle of side L , start moving with a constant tangential acceleration towards each other in a cyclic manner, forming spiral loci that coverage at the centroid of the triangle. The length of one such spiral locus will be
 (A) $L/3$ (B) $2L/\sqrt{3}$ (C) $L/\sqrt{2}$ (D) $2L/3$.

ONLY ONE OPTION IS CORRECT.

Q.1	C	Q.2	C	Q.3	C	Q.4	B	Q.5	B	Q.6	B	Q.7	C
Q.8	B	Q.9	B	Q.10	A	Q.11	D	Q.12	D	Q.13	C	Q.14	A
Q.15	B	Q.16	B	Q.17	D	Q.18	C	Q.19	C	Q.20	B	Q.21	C
Q.22	B	Q.23	B	Q.24	A	Q.25	A	Q.26	C	Q.27	B	Q.28	C
Q.29	A	Q.30	B	Q.31	C	Q.32	B	Q.33	B	Q.34	D	Q.35	A
Q.36	C	Q.37	B	Q.38	C	Q.39	A	Q.40	A	Q.41	A	Q.42	B
Q.43	C	Q.44	C	Q.45	C	Q.46	B	Q.47	C	Q.48	C	Q.49	C
Q.50	B	Q.51	C	Q.52	C	Q.53	D	Q.54	B	Q.55	D	Q.56	C
Q.57	C	Q.58	D	Q.59	A	Q.60	D	Q.61	D	Q.62	D	Q.63	A
Q.64	B	Q.65	B	Q.66	B	Q.67	C	Q.68	D	Q.69	B	Q.70	A
Q.71	C	Q.72	B	Q.73	A	Q.74	B	Q.75	C	Q.76	A	Q.77	D
Q.78	A	Q.79	B	Q.80	A	Q.81	B	Q.82	B	Q.83	B	Q.84	B
Q.85	C	Q.86	B	Q.87	B	Q.88	A	Q.89	B	Q.90	D		

KINEMATICS

- ✓ 1. A particle is thrown up with a certain velocity and at an angle θ with the horizontal. Plot the variation of kinetic energy with time.
- ✓ 2. A point traveling along a straight line traversed $1/3$ of the distance with velocity v_0 . The remaining part of the distance was covered with velocity v_1 for half time and velocity v_2 for the other half of the time. Find the mean velocity of the point average over the whole time of motion. *(grodan)*
3. A swimmer crosses a flowing stream of width 'w' to and fro in time t_1 . The time taken to cover the same distance up and down the stream is t_2 . If t_3 is the time swimmer would take to swim a distance $2w$ in still water prove that $t_1^2 = t_2 t_3$.
- ✓ 4. A point moves in xy-plane according to equation $x = at$, $y = at(1 - bt)$ where a and b are positive constants and t is time. Find the instant at which velocity vector is at $\pi/4$ with acceleration vector.
- ✓ 5. A stone is released from an elevator going up with an acceleration a . The acceleration of the stone after the release is? *H.C. Verma*
- ✓ 6. A body is thrown up from a lift with velocity u relative to lift. If its time of flight with respect to lift is t then acceleration of the lift is?
- ✓ 7. A point moves rectilinearly. Its displacement x at time t is given by $x^2 = t^2 + 1$. Find its acceleration at time t .
- ✓ 8. A particle has an initial velocity of 9 m/s due east and constant acceleration of 2 m/s^2 due west. The distance covered by particle in the 5th second of its motion is?
- ✓ 9. A open lift is coming down from the top of a building at a constant speed $v = 10 \text{ m/s}$. A boy standing on the lift thrown a stone vertically upwards at a speed of 30 m/s w.r.t. himself. Find the time after which he will catch the stone.
- ✓ 10. At a height of 0.4 m from the ground the velocity of projectile in vector form is $\vec{v} = (6\hat{i} + 2\hat{j})$ (the x-axis is horizontal and y-axis is vertically upwards). Find the angle of projection of the projectile.
11. Three particles start from origin at the same time. One with velocity v_1 along positive x - axis, the second along the positive y - axis with a velocity v_2 and the third along the line $y=x$ with such a speed that all the three always stay in straight line. Find the value of v_3 .
- ✓ 12. Two particles start moving from the same point along the same straight line. The first moves with constant velocity ' v ' and the second with constant acceleration ' a '. During the time that elapses before the second catch the first, find the greatest distance between the particles and the time in which second particle catches the first.
13. A ball is thrown up with a certain velocity at angle θ to the horizontal. Plot the variation of kinetic energy with height of the particle.
- ✓ 14. Two cars start off to race with velocity 4 m/s and 2 m/s and travel in straight line with uniform acceleration 1 m/s^2 and 2 m/s^2 respectively. If they reach the final point at same instant, then find the length of the path.
- ✓ 15. The velocity of a particle varies with time as shown below. The distance travelled by the particle during $t=2\text{s}$ and $t=6\text{s}$ is?



$$\frac{4\omega^2 n^2}{(n^2 - 1)n \times v^2}$$

PHYSICS

16. The position vector of a particle varies with time as $\vec{r} = \vec{r}_0 (1 - 2\alpha t)$ where $\vec{r}_0$ is a constant vector and α is a positive constant then find the distance covered by the particle during the time interval in which particle returns to its initial position. (Assume $|\vec{r}_0| = r_0$).

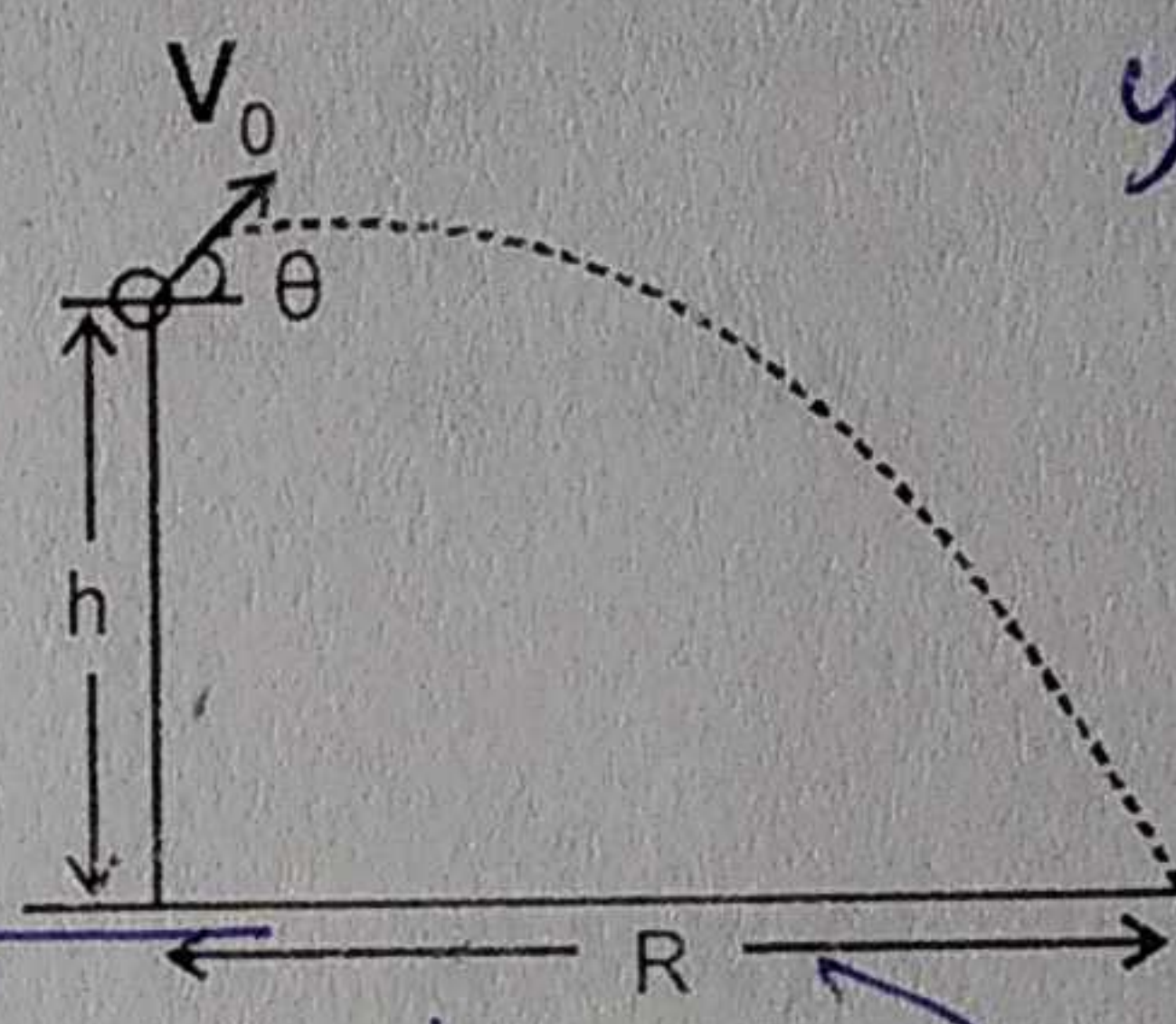
17. A train passes an observer standing on a platform. The first carriage of the train passes the observer in time $t_1 = 1s$ and the second carriage in $t_2 = 1.5s$. Find its acceleration assuming it to be constant. The length of each carriage is $l = 12m$.

18. A particle is projected from the ground at angle of 60° with horizontal speed $u = 20 m/s$. Find the radius of curvature of the path of the particle, when its velocity makes an angle of 30° with horizontal.

19. A ball is fired with an initial speed v_0 at angle θ above the horizontal from a point at height h , as shown in the figure show that the horizontal range is

$$R = \frac{v_0^2 \sin 2\theta}{2g} \left(1 + \sqrt{1 + \frac{2gh}{v_0^2 \sin^2 \theta}} \right)$$

$$R = \frac{v_0^2 \sin 2\theta}{2g} \left(1 + \sqrt{1 + \frac{2gh}{v_0^2 \sin^2 \theta}} \right)$$

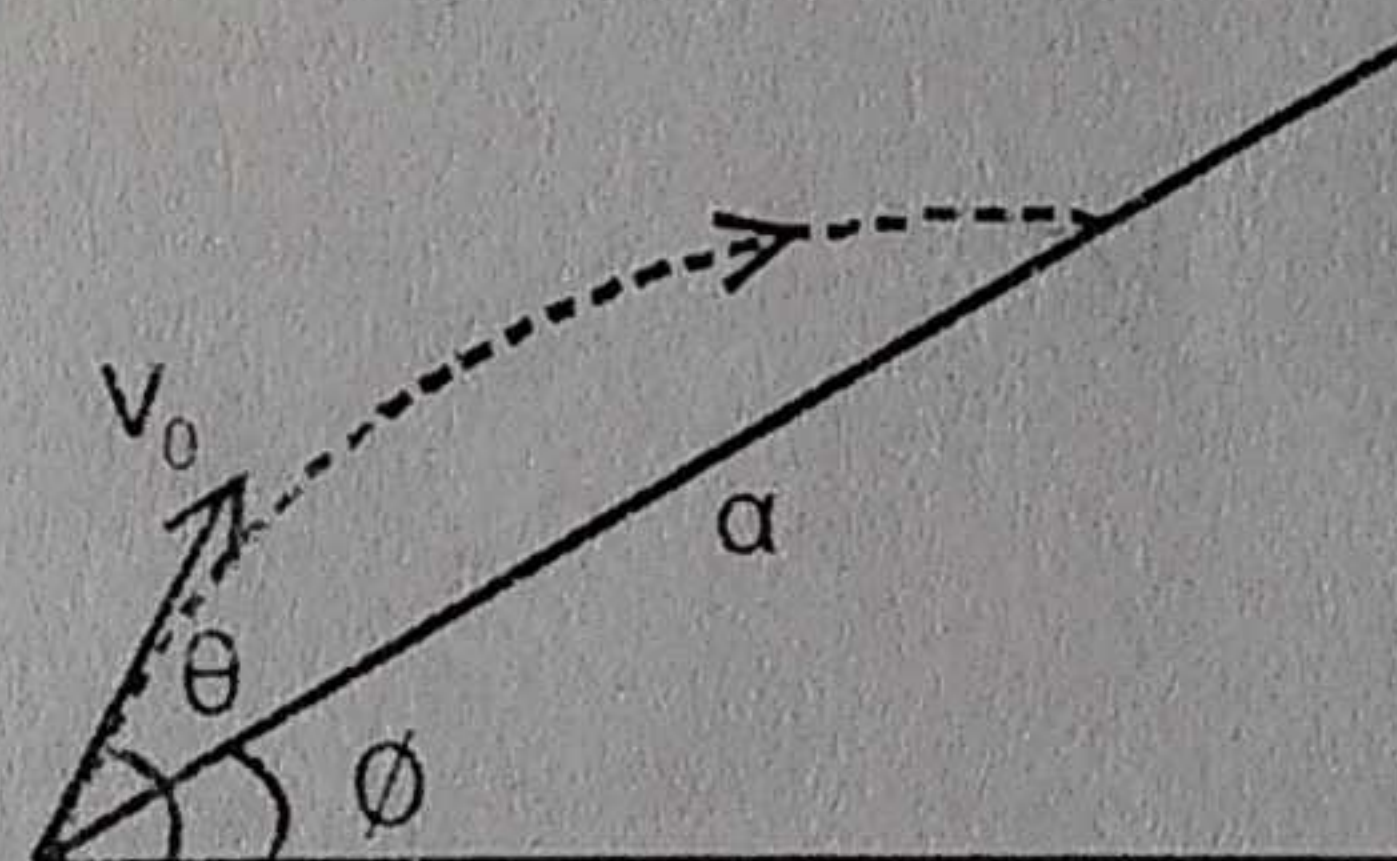


$$y = v_0 \sin \theta t - \frac{gt^2}{2}$$

$$y = -h$$

20. A projectile is fired up an incline (incline angle ϕ) w.r.t. the horizontal ($\theta > \phi$) as shown in figure. Show that the projectile travels a distance α up the incline, where

$$d = \frac{2v_0^2 \cos \theta \sin(\theta - \phi)}{g \cos^2 \phi}$$



21. Two particles are projected from the ground one at an angle α below 45° and other at α above 45° . They land at the level from which they were fired. Show that the difference in their time of flight to

$$T_2 - T_1 = \frac{2\sqrt{2}v_0 \sin \alpha}{g}$$

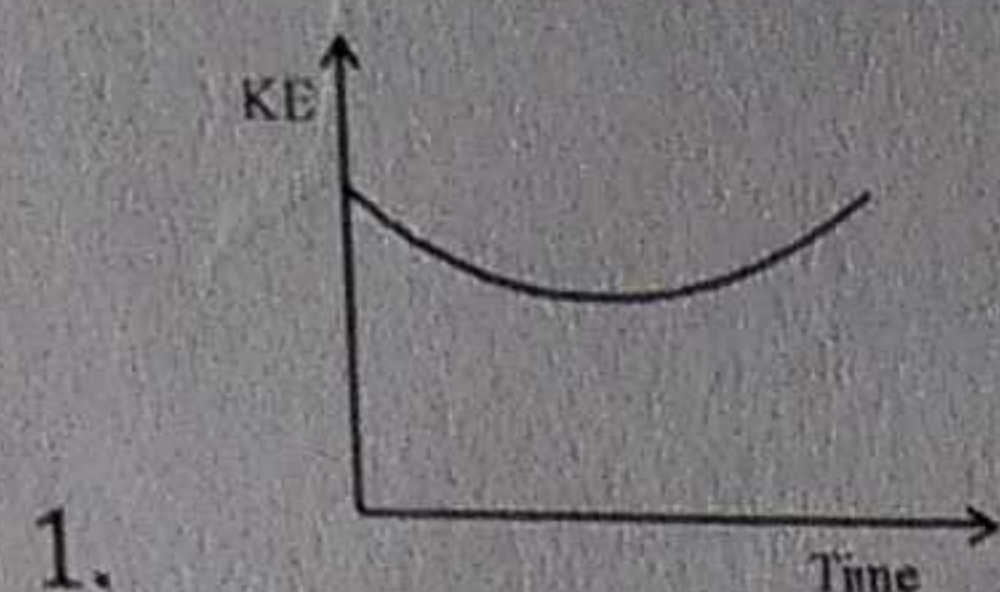
22. A boy ^{throws} shows a ball with velocity $20m/s$ from a $10m$ high balcony, directed at a target on the ground $40m$ from the base of the building. By what horizontal distance does the ball miss?

23. A particle moves with constant acceleration a in the positive x - axis. At $t=0$ the particle is at origin is at rest, then plot the graph between $(\text{velocity})^2$ and displacement.

24. A ball is dropped from the top of a building. The ball takes $0.5 s$ to fall the $3m$ length of a window some distance from the top of the building. How fast was the ball going as it passes the top of the window? How far is to the top of the window from the point at which ball was dropped? (Take $g = 9.8 m/s^2$)

25. A body moving with a constant retardation in straight line travels $5.7 m$ and $3.9 m$ in 6^{th} and 9^{th} second, respectively. When will the body come momentarily to rest and what is acceleration of the body.

Key



2. $\frac{3v_0(v_1 + v_2)}{v_1 + v_2 + 4v_0}$

3. To Prove

4. $\frac{1}{b}$

5. g downward

6. $\left(\frac{2u - gt}{t}\right)$ upwards

7. $\frac{1}{x^3}$

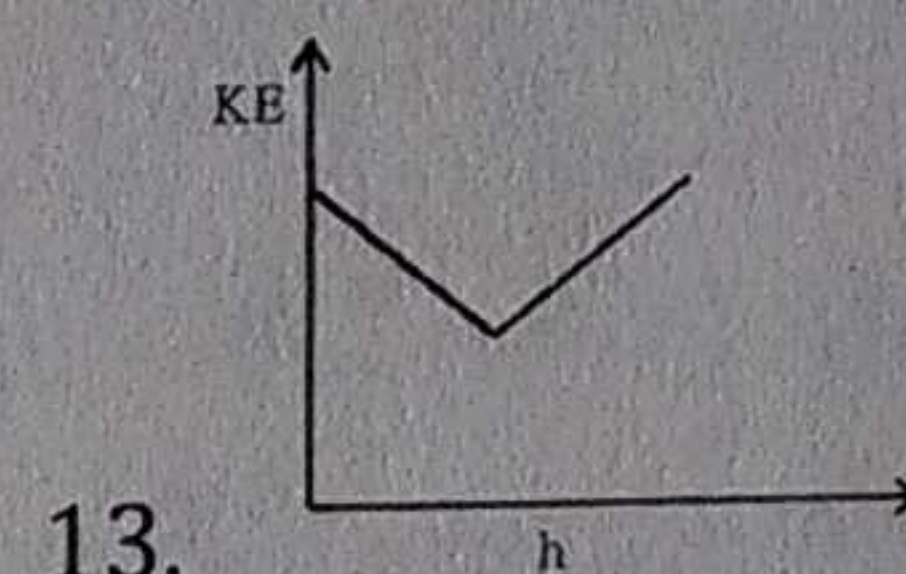
8. 0.5 m

9. 6s

10. 30°

11. $\frac{v_1 v_2 \sqrt{2}}{v_1 + v_2}$

12. $\frac{v^2}{2a}, \frac{v}{a}$



14. 24m

15. $(2\pi + 40)m$

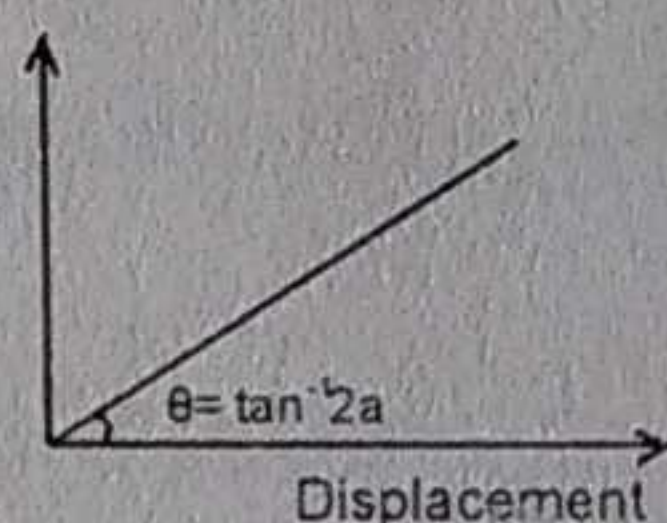
16. $r_0/2\alpha$

17. -3.2 m/s²

18. 15.4 m

19. To Prove

20. To Prove



21. To prove

22. 20.2m

23.

24. 0.64 m

25. -0.6 m/s², 15 sec